

Sharp Stability Bounds for Graph Convolutional Neural Networks

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Abstract—Graph convolutional neural networks (GCNNs) are powerful models for processing and learning from graph data. The inference-time stability of a GCNN quantifies how much its output changes at most when edges and graph signals are perturbed, assessing its robustness to temporal variations and adversarial attacks. However, existing stability bounds for GCNNs typically hold for small perturbations, first-order graph convolution, or a single layer. In this work, we show that the stability of a GCNN under signal and structure perturbations can be characterized by the Lipschitz constants of the layers in a general setting, and derive Lipschitz constant bounds for groups of graph convolutional filters that lead to non-asymptotic GCNN stability bounds that scale with the spectral norm of the convolutional coefficients and do not always grow with feature dimension. The analysis covers layer biases, graph-level architectures, and node injection. Many of our bounds generalize or improve upon existing ones, while the bounds derived from various approaches each dominate in tightness in some regime, which is confirmed by numerical experiments on both synthetic graph filters and common GCNNs trained on real-world datasets.

Index Terms—Graph neural networks, stability analysis, graph signal processing.

I. INTRODUCTION

Processing and learning from graph data have received increasing attention in the past decade. Graph signal processing has emerged as a framework for processing signals supported on graphs by leveraging their structural information, extending concepts such as convolution, Fourier transform, and filters from Euclidean spaces to the space of graph signals [1], [2]. A central tool in graph signal processing is the graph convolutional filter (GCF), defined as a polynomial in the graph shift operator, a linear operator associated with the graph that propagates the signal from each node to its neighborhood. Combining local aggregation with nonlinearity, the graph convolutional neural network (GCNN) defines a class of neural network models where graph signals are processed by banks of GCFs and element-wise activation functions across layers [3], [4]. GCNN models have achieved great success in classification and regression tasks on graphs, in fields such as quantum chemistry [5], wireless communications [6], recommendation systems [7], and document classification [8].

In real-world applications, the input graph structure and signals of a GCNN are susceptible to perturbations at inference time, such as uncertainty and noise in graph construction [9], [10], [11], adversarial attacks [12], [13], and temporal variations [14], [15]. In any case, a well-behaved GCNN should maintain a stable output when the inputs change slightly. However, adversarial examples with very small deviations from the

original input can produce vastly different outputs, enabling an imperceptible attack to degrade the overall performance of a GCNN [12], [16], [17]. Therefore, bounding the deviation of a GCNN's output embedding under structure and signal perturbations has been an active area of research, aimed at understanding why GCNNs are vulnerable and developing training strategies to enhance robustness [18], [19], [20], [21].

Technically, a stability bound is usually stated as a finite upper bound on the Lipschitz constant of a GCNN with respect to a distance in a feature or graph-structure domain, since GCNNs with smaller Lipschitz constants (relative to their embedding magnitude or classification margins) are more stable. Because a stability bound characterizes the worst-case output perturbation for a fixed magnitude of input perturbation, it can be viewed as a form of guarantee on prediction robustness. For example, in classification tasks, one can compute a robustness certificate for a graph example by dividing the classification margin by an upper bound of the Lipschitz constant in the perturbation domain [22], [23]. The certification radius increases as the bounds tighten, motivating improvements to the bounds. More importantly, a more precise bound also helps characterize the dependent factors of GCNN stability, thereby providing better theoretical grounding for designing architectures and algorithms to improve stability.

There are several approaches to bounding the Lipschitz constant of a GCNN under structure or signal perturbations. For GCNNs with only first-order graph convolutions, one can apply norm inequalities directly on the difference of embedding functions layer-by-layer, resulting in bounds that depend on the product of the spectral norms of the weight matrices [24], [23]. The case for signal perturbations was improved by leveraging spectral graph theory [20]. On the other hand, because a GCNN is built primarily from GCFs, its stability depends crucially on that of the underlying GCFs. In fact, stability bounds for GCNNs can be obtained through bounds for GCFs [25], [26], reducing the analysis for the full network in higher orders to the elementary case of a single filter. The Lipschitz constant over signal perturbations is trivially the spectral norm of the filter operator. For structure perturbations, stability bounds for both absolute and relative perturbations to the graph shift operator were derived by [25], in which higher-order polynomial terms in the difference are collected into an asymptotic term, since the linear term dominates them in the small-perturbation limit. The binomial expansion approach was also taken by [27], accounting for changes in eigenstructure. The strict linear stability of GCFs was established in [28] and subsequently improved, while being extended to low-pass filters in [29].

The structure stability bound for a GCNN that depends on bounds for single GCFs [25, Theorem 4] scales linearly with the product of feature dimensions of all layers. The same approach applied to generalizations of the GCNN, including graphon neural networks [30], manifold neural networks [31], and algebraic neural networks [26], results in bounds with the same scaling. Such a dependence is unfavorable, compared with results for first-order GCNNs, where the spectral norms do not necessarily scale with feature dimension. Notably, the stability of a restricted version of GCNN allowing arbitrary orders of graph convolution was bounded in [32], without explicit dimension dependence. However, a dimensionless structural stability bound for arbitrary GCNNs is still absent.

In this work, we propose a framework for analyzing the inference-time stability of GCNNs systematically and derive concrete stability bounds that rely on minimal assumptions, many of which generalize or improve over existing ones. The central idea is to consider all GCFs in a GCNN layer as a single matrix-to-matrix function, called a matrix convolutional graph filter (mGCF), so that the bounds do not directly scale with feature dimension. The stability analysis is performed by examining the polynomial structure of mGCFs, as well as their spectral properties in the Fourier domain. Our contribution can be summarized as follows:

- By purely considering the function-composition structure, we show that the output deviation of a GCNN under signal and structure perturbations can be bounded by the (uniform) partial Lipschitz constants of its layers and the norms of the bias terms, and extend the results to asymptotic bounds and graph-level readout structures.
- We derive an exact expression for the signal Lipschitz constant and upper bounds on the structure Lipschitz constant of mGCFs via three distinct approaches: bounding by matrix norm inequalities, deriving asymptotic bounds through binomial expansion, and generalizing to filters with arbitrary frequency response and studying their spectral properties, all of which resulted in dimensionless bounds.
- We show that the proposed structure Lipschitz bounds for mGCFs are tight and complement one another both theoretically and empirically; in particular, a lower bound matches the upper bounds for certain cases, up to a constant, or asymptotically.

Related Work. Graph neural networks (GNNs) usually refer to neural network architectures for learning on graphs in a sense broader than GCNNs. The structure stability of aggregate-GNNs, a class of GNNs with node-varying convolution, was studied in [33]. There are also GNNs that use various message-passing mechanisms without a clear convolutional structure, such as self-attention [34], nonlinear aggregations [35], and positional encodings [36]. The stability of some of these components has been studied [37], [38]; however, full structure stability bounds for general message-passing GNNs are still an open problem. There is also a parallel line of studies on the stability of GCNNs in probabilistic settings, where the bounds reflect the average change in the output over the expectation of certain random signals

or GSO perturbation models [39], [40], [41], [42], [43]. The probabilistic bounds account for the data distribution and match empirical observations well. However, they do not reflect the worst-case scenario and can only provide robustness certificates that hold probabilistically. Therefore, probabilistic and deterministic bounds are different perspectives on the same phenomenon and complement each other. In particular, because a bound in expectation is only meaningful when it is tighter than deterministic bounds for the worst-case in a distribution, our bounds serve as a strong baseline. Other notions of GCNNs were also studied, such as the algorithmic stability of the trained weights and the stability of the loss function with respect to the weight matrices, often in the context of generalization.

Notations. $\mathbb{R}$ denotes the set of real numbers. Bold upper-case letters $\mathbf{A} \in \mathbb{R}^{n \times m}$ and lowercase letters $\mathbf{x} \in \mathbb{R}^n$ denote matrices and vectors respectively, whose entries are $[\mathbf{A}]_{ij}$ and $[\mathbf{x}]_i$. Uppercase letters in calligraphic font such as $\mathcal{V}, \mathcal{E}, \mathcal{D}, \mathcal{K}$ are sets, while $\mathcal{H}$ is reserved for order-4 tensors. The transpose of a matrix (or a vector, seen as a column vector) is denoted by $\cdot^T$. $\text{diag}(\mathbf{x})$ denotes a diagonal matrix whose diagonal is formed by the vector $\mathbf{x}$. Let $\mathbf{A}_{i,:}$ and $\mathbf{A}_{:,j}$ denote the i^{th} row vector and j^{th} column vector of a matrix $\mathbf{A}$, respectively. The notation $\langle \cdot, \cdot \rangle$ denotes the tensor contraction product $[\langle \mathbf{X}, \mathcal{H} \rangle]_{kl} = \sum_{ij} [\mathbf{X}]_{ij} [\mathcal{H}]_{ijkl}$. For a condition p , $\mathbb{1}_p$ equals 1 when p is true and equals 0 otherwise. $\mathbf{1}$ denotes a vector composed of ones. Given an integer $n \in \{0, 1, 2, \dots\}$, $[n]$ is the set $\{1, 2, \dots, n\}$. The notations $\|\cdot\|_2$, $\|\cdot\|_F$ and $\|\cdot\|_{\max}$ denote the spectral norm, Frobenius norm, and element-wise max norm of a matrix, respectively. The symbol $:=$ means definition. $\mathcal{U}[a, b]$ denotes a uniform distribution on $[a, b]$, and $\mathcal{N}(\mu, \sigma^2)$ is a Gaussian distribution with mean μ and variance σ^2 .

II. GRAPH CONVOLUTIONAL NEURAL NETWORKS

In this section, we formally introduce GCNN and related objects in graph signal processing that are crucial for analyzing GCNNs. We define a graph as a set of nodes $\mathcal{V} = \{1, \dots, N\}$ and edges $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ on it. The neighborhood of a node i is $\mathcal{N}_i = \{j : (i, j) \in \mathcal{E}\}$. A graph is undirected if for every $(i, j) \in \mathcal{E}$, (j, i) is also in $\mathcal{E}$. A graph shift operator (GSO) related to a graph $(\mathcal{V}, \mathcal{E})$ is any matrix $\mathbf{S} \in \mathbb{R}^{N \times N}$ satisfying

$$i \neq j, (i, j) \notin \mathcal{E} \Rightarrow [\mathbf{S}]_{ij} = 0.$$

Because whenever $[\mathbf{S}]_{ij} \neq 0, i \neq j$, there is $(i, j) \in \mathcal{E}$, a GSO encodes the structure of a graph. Examples of GSOs include the adjacency matrix $[\mathbf{A}]_{ij} = \mathbb{1}_{(i,j) \in \mathcal{E}}$, the graph Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ (where $\mathbf{D} = \text{diag}(\mathbf{A}\mathbf{1})$ is the diagonal degree matrix), and the symmetrically normalized graph Laplacian $\tilde{\mathbf{L}} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$, among others. The GSO can also be the weighted adjacency matrix or Laplacians for graphs with scalar edge weights.

A graph signal on $\mathcal{V}$ is a vector $\mathbf{x} \in \mathbb{R}^N$, where $[\mathbf{x}]_i$ is the signal for node i . The graph shift $\mathbf{x} \mapsto \mathbf{S}\mathbf{x}$ describes the process of applying a GSO (by left-multiplying) to a graph signal, propagating the signal on each node to only the neighboring nodes due to the sparsity of $\mathbf{S}$. A graph

convolutional filter (GCF) propagates a graph signal $\mathbf{x} \in \mathbb{R}^N$ farther by repeatedly applying the graph shift, written as

$$\mathbf{y} = \sum_{k=0}^K \theta_k \mathbf{S}^k \mathbf{x}, \quad (1)$$

where $\{\theta_k\}_{k=0}^K$ are the convolutional coefficients. We call k and K the *orders* of graph shifts and the graph convolution, respectively. A GCF of order K propagates signals within a radius of K hops. The GCF admits a convolution theorem through the graph Fourier transform (GFT). Let $\mathbf{S} = \mathbf{V}\mathbf{J}\mathbf{V}^{-1}$ be an eigendecomposition of the GSO. The GFT and its inverse are defined as the linear operators $\mathbf{V}^{-1}$ and $\mathbf{V}$ (which are unique up to the choice of eigenbasis, for a fixed eigenvalue order), and the eigenvalues of $\mathbf{S}$ are called ‘frequencies’. For spectral analysis in this work, we assume $\mathbf{S}$ to be symmetric. In such a case, there is always an eigendecomposition $\mathbf{S} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_N)$ is the diagonal matrix consisting of the eigenvalues, and $\mathbf{V}$ is an orthogonal matrix forming an eigenbasis, so the GFT is given by $\mathbf{V}^T$. A graph filter can be defined by a frequency response $h : \mathbb{R} \rightarrow \mathbb{R}$ as

$$\mathbf{y} = \mathbf{V} \text{diag}(h(\lambda_1), \dots, h(\lambda_N)) \mathbf{V}^T \mathbf{x}. \quad (2)$$

The GCF (1) is then defined by the polynomial frequency response $h(\lambda) = \sum_{k=0}^K \lambda^k \theta_k$. From this perspective, a GCF is an element-wise product between the signal in the frequency domain and the frequency response.

The GCNN is a class of deep learning models on graphs using banks of GCFs for signal propagation. Formally, we write the node embedding function of a GCNN as

$$\Phi(\mathbf{X}, \mathbf{S}) = f^{(L)}(\sigma[\dots f^{(2)}(\sigma[f^{(1)}(\mathbf{X}, \mathbf{S})], \mathbf{S}) \dots], \mathbf{S}), \quad (3a)$$

$$\text{with } f^{(\ell)}(\mathbf{X}, \mathbf{S}) = \sum_{k \in \mathcal{K}_\ell} \mathbf{S}^k \mathbf{X} \mathbf{\Theta}_k^{(\ell)} + \mathbf{1} \mathbf{b}^{(\ell)T}, \quad (3b)$$

where $\mathbf{X} \in \mathbb{R}^{N \times F_0}$, $\mathbf{S} \in \mathbb{R}^{N \times N}$ are the node feature matrix and the GSO (of a certain type) associated with a graph, σ is the activation function defined as applying $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ element-wise, $\mathcal{K}_\ell \subseteq \{0, 1, 2, \dots\}$ specifies the parameterized graph shift orders, and $\mathbf{\Theta}_k^{(\ell)}$, $\mathbf{b}^{(\ell)}$ are the learnable weight matrices and layer biases.

Common GCNN layers are listed in Table I. ChebNet [44] parameterizes the GCFs with the Chebyshev polynomials, and is among the first GCNNs. GCN simplifies the polynomial to a linear function, uses a ‘renormalized’ Laplacian, and is further simplified by SGC [45] by collapsing all the layers into an order- K graph shift. GraphSAGE [35] was proposed for inductive graph learning. Using the mean aggregator, GraphSAGE is also a GCNN with an asymmetric GSO. LightGCN [7] and GraphHeat [46] use restricted parameterizations of order- K GCFs to reduce computation load while retaining long-range signal propagation. Such a distinction from GCNNs that allow all values of parameterized $\mathbf{\Theta}_k$, $k \in \mathcal{K}$ is indicated in the last column ‘Full Param’. Some GNNs have multilayer perceptrons (MLPs) before or after the convolution component. We separate the MLP and decompose it into zero-order GCNN

layers, so the full network is a GCNN with varying $\mathcal{K}_\ell$ [4]. For example, let $f_\theta(\mathbf{x}) = \sigma(\mathbf{W}_2 \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2)$ be a two layer MLP applied to all node features $\mathbf{X}_i$. Then, it can be written as two GCNN layers with $\mathcal{K} = \{0\}$ and $\mathbf{\Theta}_0 = \mathbf{W}_1^T$ or $\mathbf{W}_2^T$. Examples include the GIN [47], ChebNetII [48], and APPNP [49].

III. GCNN STABILITY THROUGH LIPSCHITZ CONTINUITY

A function is Lipschitz continuous if the output deviation can be linearly bounded by the input change, under suitable metrics. In other words, the supremum of the ratio between output and input differences, called the Lipschitz constant, is a finite value. Let $f : \mathbb{R}^{a \times b} \rightarrow \mathbb{R}^{c \times d}$ be any function (which includes the cases with vector or scalar domains); its Lipschitz constant can be defined using the Frobenius norm as

$$\text{Lip}(f) := \sup_{\mathbf{X} \neq \hat{\mathbf{X}}} \frac{\|f(\mathbf{X}) - f(\hat{\mathbf{X}})\|_F}{\|\mathbf{X} - \hat{\mathbf{X}}\|_F} \in [0, \infty].$$

In this section, we will bound the Lipschitz constants of a GCNN (3a) for both the signal input $\mathbf{X}$ and the GSO input $\mathbf{S}$ in terms of the (partial) Lipschitz constants of its layer functions, and extend the results to graph-level predictions. We require the bound to be independent of the inputs and hold for a fixed number of nodes N . These restrictions will be discussed in detail in Section V in the context of concrete bounds.

A. Lipschitz Constants

The GCNN node embedding function (3) and the layer functions take two inputs, so their stability has been studied for each variable with the other fixed. In light of existing stability bounds, we formulate the stability of a GCNN or a layer through the Lipschitz continuity of matrix functions, in order to derive general stability bounds that do not contain feature dimension terms. Let $f(\mathbf{X}, \mathbf{S})$ be a function on $\mathbb{R}^{N \times d} \times \mathcal{S}$, where $\mathcal{S}$ can be any subset of $\mathbb{R}^{N \times N}$, representing the domain of GSOs. We define Lipschitz constants in both the signal variable $\mathbf{X}$ and the GSO variable $\mathbf{S}$ symmetrically:

$$\begin{cases} \text{Lip}(f; \mathbf{S}) := \sup_{\mathbf{X} \neq \hat{\mathbf{X}}} \frac{\|f(\mathbf{X}, \mathbf{S}) - f(\hat{\mathbf{X}}, \mathbf{S})\|_F}{\|\mathbf{X} - \hat{\mathbf{X}}\|_F}, \\ \text{Lip}(f; \mathbf{X}) := \sup_{\mathbf{S} \neq \hat{\mathbf{S}}} \frac{\|f(\mathbf{X}, \mathbf{S}) - f(\mathbf{X}, \hat{\mathbf{S}})\|_F}{\|\mathbf{S} - \hat{\mathbf{S}}\|_2}. \end{cases} \quad (4)$$

The definition in (4) corresponds to the (smallest) Lipschitz constants of the partially evaluated functions $f(\cdot, \mathbf{S})$ and $f(\mathbf{X}, \cdot)$ [50], so we will call them *signal Lipschitz constant* and *structure Lipschitz constant*, respectively.

B. A General Stability Bound

To bound the difference between the outputs of a GCNN for different signal and GSO inputs, we need uniform bounds on the partial Lipschitz constants of the mGCFs; that is, the fixed variables in (4) can take any value in their domains. The reason is as follows: For signals, an intermediate signal matrix $\mathbf{X}^{(\ell)}$ can take many values in the space $\mathbb{R}^{N \times F_\ell}$; while for GSOs, each layer function takes the same input as the whole network,

Name	GSO	$\bar{\rho}$	$\mathcal{K}$	Full Param.
GCN [8]	$\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2}$	1	$\{1\}$	Yes
GraphSAGE-mean [35]	$\tilde{\mathbf{D}}^{-1} \tilde{\mathbf{A}}$	1	$\{1\}$	Yes
SGC [45]	$\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2}$	1	$\{K\}$	Yes
ChebNet [44]	$2\hat{\mathbf{L}}/\rho(\hat{\mathbf{L}}) - \mathbf{I}$	1	$\{0, 1, \dots, K\}$	Yes
GraphHeat [46]	$\hat{\mathbf{L}}$	2	$\{0, 1, \dots, K\}$	No
LightGCN [7]	$\mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$	1	$\{0, 1, \dots, K\}$	No
Stable-ChebNet	$\hat{\mathbf{L}}$	2	$\{0, 1, \dots, K\}$	No
Node-feature MLP	any	any	$\{0\}$	Yes
GIN with the first layer of MLP [47]	$\mathbf{A} + (1 + \epsilon)\mathbf{I}$	$N + \epsilon$	$\{1\}$	Yes
ChebNetII [48]	$2\hat{\mathbf{L}}/\rho(\hat{\mathbf{L}}) - \mathbf{I}$	1	$\{0, 1, \dots, K\}$	No
GPR-GNN	$\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2}$	1	$\{0, 1, \dots, K\}$	No

TABLE I: Specifications of common GCNN layers. $\bar{\rho}$ denotes the GSO’s maximal spectral radius. ‘Full Param.’ indicates whether the model’s parameterization allows all possible values of $\{\Theta_k : k \in \mathcal{K}\}$. $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, $\tilde{\mathbf{D}} = \text{diag}(\tilde{\mathbf{A}}\mathbf{1})$ are the augmented adjacency and degree matrices. Because APPNP [49] is a special case of GPR-GNN, it is not listed in a separate entry.

but we require the bound to be independent of the inputs, so we also need bounds uniform in $\mathcal{S}$. Because $\mathcal{S}$ is a bounded domain, we obtain a uniform bound of $\text{Lip}(f; \mathbf{S})$ by simply taking the supremum over $\mathcal{S}$. For $\text{Lip}(f; \mathbf{X})$, because existing structure stability bounds for GCFs often scale linearly with the norm of the input, we choose to normalize by $\|\mathbf{X}\|_{\text{F}}$:¹

$$\overline{\text{Lip}}(f) := \sup_{\mathbf{X} \neq 0} \frac{\text{Lip}(f; \mathbf{X})}{\|\mathbf{X}\|_{\text{F}}}. \quad (5)$$

With the Lipschitz constants defined for functions of graph signal matrices, we can derive a general stability bound for GCNNs depending on the stability of the layer functions.

Theorem 1: Let Φ be a GCNN embedding function (3a) where the scalar activation function satisfies $\sigma(0) = 0$ and $\text{Lip}(\sigma) \leq B_\sigma$. Suppose for each $\ell \in [L]$, the layer function $f^{(\ell)}$ satisfies

$$\begin{cases} \overline{\text{Lip}}(f^{(\ell)}) \leq C_\ell < \infty, \\ \text{Lip}(f^{(\ell)}; \mathbf{S}) \leq B_\ell < \infty, \forall \mathbf{S} \in \mathcal{S} \end{cases} \quad (6)$$

then for any $\mathbf{X}, \hat{\mathbf{X}} \in \mathbb{R}^{N \times F_0}$; $\mathbf{S}, \hat{\mathbf{S}} \in \mathcal{S}$,

$$\begin{aligned} \left\| \Phi(\mathbf{X}, \mathbf{S}) - \Phi(\hat{\mathbf{X}}, \hat{\mathbf{S}}) \right\|_{\text{F}} &\leq B_\sigma^{L-1} B_1 B_2 \cdots B_L \\ &\left(\epsilon_X + \epsilon_S \sum_{\ell=1}^L \frac{C_\ell}{B_\ell} \left(\|\mathbf{X}\|_{\text{F}} + \sum_{k=1}^{\ell-1} \frac{\sqrt{N} \|\mathbf{b}^{(k)}\|_2}{B_1 \cdots B_k B_\sigma^{k-1}} \right) \right), \end{aligned} \quad (7)$$

where $\epsilon_X = \|\mathbf{X} - \hat{\mathbf{X}}\|_{\text{F}}$, $\epsilon_S = \|\mathbf{S} - \hat{\mathbf{S}}\|_2$ are the perturbation sizes.

Proof: Please refer to Section S1-A of the supplemental material. ■

¹When f is linear in $\mathbf{X}$, we can rewrite (5) as

$$\overline{\text{Lip}}(f) = \sup_{\mathbf{S} \neq \hat{\mathbf{S}}} \frac{\left\| f(\cdot, \mathbf{S}) - f(\cdot, \hat{\mathbf{S}}) \right\|_{\text{op}}}{\|\mathbf{S} - \hat{\mathbf{S}}\|_2},$$

where $\|\mathcal{A}\|_{\text{op}} := \sup_{\mathbf{X} \neq 0} \|\mathcal{A}\mathbf{X}\|_{\text{F}} / \|\mathbf{X}\|_{\text{F}}$ is the operator norm. This alternative expression takes the form of a Lipschitz constant and matches the notion of linear stability of graph filters in [29].

Theorem 1 shows that the stability of a multi-layer GCNN to both inputs is controlled by the partial Lipschitz bounds of the layer functions uniform on the domains. The signal Lipschitz constant bounds B_ℓ as well as B_σ appear in the bound as product terms, while the structure bounds C_ℓ enter as a (weighted) sum, reflecting the different roles of signal and structure inputs in (3). The structure sensitivity (the coefficient on ϵ_S) scales with the norm of the input signal $\|\mathbf{X}\|_{\text{F}}$, as well as the norms of the bias vectors, scaled by the number of nodes and the signal Lipschitz constants of successive layers—the denominators on the right-hand side of (7) cancel out the part of $B_1 \cdots B_L$ and B_σ^{L-1} for preceding layers. When the bias terms $\mathbf{b}^{(k)}$ on each layer are zero, (7) yields a uniform bound over $\mathbf{X}$:

$$\overline{\text{Lip}}(\Phi) \leq B_\sigma^{L-1} \left(\prod_{\ell=1}^L B_\ell \right) \sum_{\ell=1}^L \frac{C_\ell}{B_\ell}, \quad (8)$$

but such a bound does not exist with nonzero bias terms. This is a key difference between the structure Lipschitz constants of a GCNN and an mGCF: even when the network input $\mathbf{X}^{(0)}$ is zero, an intermediate signal matrix $\mathbf{X}^{(\ell)}$, $\ell \geq 1$, being the input to $f^{(\ell+1)}$, may not be zero because of the bias, and normalization with $\|\mathbf{X}\|_{\text{F}}$ diverges at 0.

Remark 1.1 (Extension to graph distance metrics): Theorem 1 naturally holds in a more general setting with essentially the same proof. First, the GSO distance $d_S(\mathbf{S}, \hat{\mathbf{S}}) = \|\mathbf{S} - \hat{\mathbf{S}}\|_2$ can be replaced by any semimetric d on $\mathcal{S}$ (that is, a metric, but the triangle inequality need not hold). For example, some studies [51], [52] consider the relative distance between GSOs, which is a semimetric on some domains but is ill-defined in others (see discussions in Appendix A). Other choices of d include any matrix norm on the GSO difference, graph-editing distances [53], [54], and the tree-mover’s distance [55]. In particular, [29] showed that d_S can be upper-bounded by an asymmetric graph-editing distance when the GSO is $\hat{\mathbf{L}}$, so any stability bound using d_S can be expressed in a more interpretable form.

Remark 1.2 (Extension to other GNNs): Many GNNs that are not GCNNs adopt the function-composition form (3a) and accept a range of graph inputs, with $\mathbf{S}$ replaced by the structure-edge pair $(\mathbf{A}, w)$, where $w : \mathcal{E} \rightarrow \mathcal{D}$ denotes the edge features which can encode directions and weights. Because Theorem 1 only relies on the composition of layers, it holds in this general setting if one defines a semimetric on $\mathcal{S} = \{(\mathbf{A}, w)\}$, for example, through product metrics. When no specific form of the layer functions is assumed, the term $\sqrt{N} \|\mathbf{b}^{(k)}\|_2$ should be replaced by $\sup_{\mathbf{S} \in \mathcal{S}} \|f^{(k)}(0, \mathbf{S})\|_F$.

C. Stability in Graph-level Tasks

Theorem 1 has established the stability of the node embedding function Φ of a GCNN. In node-level tasks, the i -th row of the node embedding $\Phi(\mathbf{X}, \mathbf{S})$ is treated as the prediction for node i , such as the classification logits. Therefore, the stability of the node embedding corresponds to the joint stability of all node predictions.

Next, we extend the general stability bound to GCNNs in graph-level learning tasks. In graph-level tasks, there is usually a global readout layer π that transforms the output of Φ into a single graph feature, which is further processed by a GSO-independent network κ to yield predictions for the graph. We write the graph-level GCNN architecture as

$$\varphi(\mathbf{X}, \mathbf{S}) = \kappa(\pi(\Phi(\mathbf{X}, \mathbf{S}))). \quad (9)$$

There can be more complex global readout schemes, such as reading from all intermediate signals, which we will not cover for brevity. The graph representation φ (9) appends a pooling layer π and a classifier κ to the node embeddings Φ . In addition to the extra function composition, it is helpful to also consider the structure with π and κ . The classifier κ is usually an MLP on the single graph feature and has a well-characterized Lipschitz constant. The pooling function π usually applies a permutation-invariant aggregation function $r(x_1, \dots, x_N)$ to each graph signal, most commonly mean, sum, or max [56]. We can characterize the stability of φ as follows.

Theorem 2: Let $r : \mathbb{R}^N \rightarrow \mathbb{R}$ be an aggregation function, and π be defined as $[\pi(\mathbf{X})]_p = r(\mathbf{X}_{:,p})$.

(a) Assuming all terms are finite,

$$\begin{aligned} \text{Lip}(\varphi; \mathbf{X}) &\leq \text{Lip}(\kappa) \text{Lip}(\pi) \text{Lip}(\Phi; \mathbf{X}), \\ \text{Lip}(\varphi; \mathbf{S}) &\leq \text{Lip}(\kappa) \text{Lip}(\pi) \text{Lip}(\Phi; \mathbf{S}). \end{aligned}$$

(b) The global pooling layer satisfies

$$\text{Lip}(\pi) \leq \text{Lip}(r).$$

In particular, the *sum*, *mean* and *max* pooling layers have Lipschitz constants bounded by $\sqrt{N}$, $1/\sqrt{N}$, and 1, respectively.

(c) If r is permutation-invariant, then φ is invariant under graph isomorphism.

Proof: See Section S1-B of the supplemental material. ■ Theorem 2 shows that the stability of the graph embedding function is controlled by the stability of the underlying node embedding function and the graph-level classifier, as well as

the type of aggregator used in pooling. The invariance of φ under graph isomorphism allows one to tighten the bound (7) by choosing a permutation matrix $\mathbf{P}$ so that $\mathbf{P}\mathbf{X}$ and $\mathbf{P}\mathbf{S}\mathbf{P}^T$ are closer to $\hat{\mathbf{X}}$ and $\hat{\mathbf{S}}$, respectively. However, it is generally hard to find an optimal permutation, and the improvement can be insignificant when $\hat{\mathbf{S}}$ is obtained by modifying $\mathbf{S}$.

D. Stability under Node Injection or Deletion

The general stability bound also extends to cases where new nodes are added to the graph (injection), or nodes are removed (deletion). Because the reverse process of node deletion is exactly injection, we will formalize the change in GSO in terms of node injection. Suppose there are originally N nodes in the graph represented by a GSO $\mathbf{S} \in \mathbb{R}^{N \times N}$. When M nodes are injected to the graph, we number them $N+1, \dots, N+M$ and let the new GSO be $\hat{\mathbf{S}} \in \mathbb{R}^{(N+M) \times (N+M)}$. Let the node features of the injected nodes be $\mathbf{X}_+ \in \mathbb{R}^{M \times F_0}$, then the new feature matrix is $\hat{\mathbf{X}} = [\mathbf{X}^T | \mathbf{X}_+^T]^T$. Because there are additional rows of node embeddings in the output for the injected graph, we compare the embeddings for the original N nodes. To handle the dimension mismatch, we construct an input with all injected nodes and features but no new edges, and show that the extra nodes do not affect the embeddings for the original nodes.

Proposition 3: When M nodes are added to a graph with GSO $\mathbf{S}$, the GSO becomes $\mathbf{S} \oplus \mathbf{C}$, where $\mathbf{C} = \mathbf{0}_{M \times M}$ for GSOs $\pm \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$ and $\mathbf{A}$, $\mathbf{C} = \mathbf{I}_{M \times M}$ for GSOs $\hat{\mathbf{L}}$, $\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2}$, $\mathbf{A} + \mathbf{I}$, and $\tilde{\mathbf{D}}^{-1} \tilde{\mathbf{A}}$. Moreover, for any GCNN node embedding $\Phi(\mathbf{X}, \mathbf{S})$, there is

$$\Phi(\hat{\mathbf{X}}, \mathbf{S} \oplus \mathbf{C}) = \begin{bmatrix} \Phi(\mathbf{X}, \mathbf{S}) \\ f_\theta(\mathbf{X}_+) \end{bmatrix}, \quad (10)$$

where f_θ is an MLP defined with the parameters of Φ as

$$\mathbf{X}^{(\ell)} = \sigma(\mathbf{C}\mathbf{X}^{(\ell-1)}\mathbf{Z}^{(\ell)} + \mathbf{1}\mathbf{b}^{(\ell)\top}),$$

where $\mathbf{Z}^{(\ell)} = \sum_{k \in \mathcal{K}^{(\ell)}} \Theta_k^{(\ell)}$.

Proof: Given in Section S1-D in the supplemental material. ■

Proposition 3 can be interpreted by the locality of graph convolution: a graph shift $\mathbf{x} \mapsto \mathbf{S}\mathbf{x}$ propagates the signal at a node only to its neighborhood, so the features on the extra isolated nodes cannot affect the signals on the original graph. As a result, we can bound the difference between the original and perturbed embeddings for nodes $1, \dots, N$ with the general stability bound using the intermediate input:

Corollary 3.1: When the type of GSO is among the ones listed in Proposition 3, for any GCNN node embedding function Φ and a node injection $(\mathbf{X}, \mathbf{S}) \rightarrow (\hat{\mathbf{X}}, \hat{\mathbf{S}})$,

$$\|\Phi(\mathbf{X}, \mathbf{S}) - \Phi(\hat{\mathbf{X}}, \hat{\mathbf{S}})_{1:N,:}\|_F \leq \|\Phi(\hat{\mathbf{X}}, \mathbf{S} \oplus \mathbf{C}) - \Phi(\hat{\mathbf{X}}, \hat{\mathbf{S}})\|_F.$$

Therefore, we can bound the stability of the embeddings of the original nodes for GCNNs under node injection by applying the fixed-node stability bound (7) to the augmented node features and the GSO difference between the node-augmented graph and the injected graph (with the edges further added).

E. Improving Stability in Training

Many existing GCNN stability bounds have been applied as a penalty term on the training loss, in order to enhance the stability of the trained model. However, we point out that controlling the bound itself may not guarantee robustness in some cases. Because the Lipschitz constant is linear in scaling $\text{Lip}(\alpha f) = \alpha \text{Lip}(f)$, one can obtain a model with smaller Lipschitz constants by scaling all the weight matrices by a factor $\alpha < 1$, but the magnitude of the output shrinks simultaneously. The following proposition states this fact precisely.

Proposition 4: Let Φ be a GCNN node embedding function with L layers, and $\hat{\Phi}$ be Φ with all coefficient matrices scaled by a factor $\alpha > 0$. Suppose σ is the ReLU function $\sigma(x) = x \mathbb{1}_{x>0}$, and all biases are zero. Then, for any inputs $\mathbf{X}, \mathbf{S}$, there is $\hat{\Phi}(\mathbf{X}, \mathbf{S}) = \alpha^L \Phi(\mathbf{X}, \mathbf{S})$, thus the signal and structure Lipschitz constants of $\hat{\Phi}$ are also α^L times those of Φ .

Proof: Given in Section S7 in the supplemental material. ■

The scaling effect described by Proposition 4 can be extended to the graph-level predictor φ using similar conditions. Therefore, the stability bound alone cannot predict the robustness of a GCNN. Instead, one should compare the stability bound with the magnitude of the output. When the output scale is fixed, as in node and graph regression tasks, penalizing the stability bound will not trivially shrink the output. However, the class prediction of a classification model is independent of the magnitudes of the logits, and controlling the bound may fail to improve robustness. Specifically, let $f : \mathbb{R}^n \rightarrow [C]$ be a hard-max predictor (which can be $\arg\max_i \Phi(\mathbf{X}, \mathbf{S})_{i,:}$ or $\arg\max \varphi(\mathbf{X}, \mathbf{S})$ in our case, and the domain of f is a subset of $\mathbb{R}^{N \times F_0} \times \mathcal{S}$ where alternative inputs take value), and $y = f(x_0)$ be a prediction. The robustness of the predictor f at the same x_0 is measured by the minimal perturbation $\delta_{\min}$ needed to change the predicted class of x_0 :

$$\delta_{\min} := \min\{\delta > 0 : \exists x \in \mathbb{R}^n, \|x - x_0\| \leq \delta, f(x) \neq y\}.$$

Because $\arg\max$ is independent of the scale, for any scaled version $\hat{f}$ of f , there is $f(x) = y \Leftrightarrow \hat{f}(x) = y$, so $\delta_{\min}$ is also scale-independent. Therefore, penalizing the Lipschitz bound may only shrink the Lipschitz constant, but cannot increase the robustness margin. Algorithms have been proposed to train the GCNN while yielding a lower bound to $\delta_{\min}$, called robustness certificates. We also suggest improving the robustness of GCNNs by adopting methods that consider both the Lipschitz constant and the classification margins, by leveraging our stability bounds.

IV. STABILITY OF MIMO GRAPH CONVOLUTION

Theorems 1 and 2 give general GCNN stability bounds depending on the Lipschitz constants of the layer functions. In this section, we will derive concrete signal and structure Lipschitz bounds of the layer functions computable from the coefficient matrices. Because the Lipschitz constants are not affected by bias terms, we only need to consider the convolutional part of a layer function. Stripping away the layer indices, we write

$$\gamma(\mathbf{X}, \mathbf{S}) := \sum_{k \in \mathcal{K}} \mathbf{S}^k \mathbf{X} \Theta_k, \quad (11)$$

with $\Theta_k \in \mathbb{R}^{P \times Q}$, which we call a *MIMO graph convolutional filter* (mGCF), following [57]. When the input (P) and output (Q) dimensions are both one, it reduces to an ordinary GCF. In fact, an mGCF can be written as a group of GCFs aggregated by summing at the output:

$$\gamma(\mathbf{X}, \mathbf{S})_{:q} = \sum_{p=1}^P \left(\sum_{k \in \mathcal{K}} [\Theta_k]_{pq} \mathbf{S}^k \right) \mathbf{X}_{:p}. \quad (12)$$

We choose to keep the matrix format (11) in order to analyze the group of GCFs as a whole, because otherwise the bound will factor in the feature dimensions explicitly, as demonstrated in the following proposition.

Proposition 5: If a signal map $f : \mathbb{R}^{N \times P} \times \mathcal{S} \rightarrow \mathbb{R}^{N \times Q}$ can be represented as

$$[f(\mathbf{X}, \mathbf{S})]_{:q} = \sum_{p=1}^P f_{pq}(\mathbf{X}_{:p}, \mathbf{S}), \quad (13)$$

there is

$$\begin{aligned} \text{Lip}(f; \mathbf{S}) &\leq \sqrt{PQ} \max_{p,q} \text{Lip}(f_{pq}; \mathbf{S}), \\ \overline{\text{Lip}}(f) &\leq \sqrt{PQ} \max_{p,q} \overline{\text{Lip}}(f_{pq}). \end{aligned} \quad (14)$$

Proof: See Section S1-C of the supplemental material. ■

Proposition 5 immediately yields concrete stability bounds for GCNNs by leveraging existing results: Suppose the signal and structure Lipschitz constants of the GCF with coefficients $\{[\Theta_k^{(\ell)}]_{pq}\}_{k \in \mathcal{K}}$ in a GCNN are bounded by $B_{\ell,p,q}$ and $C_{\ell,p,q}$ uniformly as in (6), then (7) holds with $B_{\ell} = \sqrt{F_{\ell} F_{\ell-1}} \max_{p,q} B_{\ell,p,q}$ and similar C_{ℓ} . However, we cannot recover through Proposition 5 the GCNN stability bounds that do not necessarily scale with feature dimension [23], [20], [32]. The dimension terms are artifacts of the proof and can make the bound vacuous for large dimensions: suppose for each layer $\mathcal{K} = \{1\}$, $\Theta_1 = \mathbf{I}_{P \times P}$, $\sigma(x) = x$, and $\mathbf{b} = 0$. Then, $\Phi(\mathbf{X}, \mathbf{I}) = \mathbf{X}$ and $\text{Lip}(\Phi; \mathbf{I}_{N \times N}) = 1$, which is independent of P . Similarly, when $P > N$ and $L = 1$, $\Phi(\mathbf{X}_0 = [\mathbf{I}_{N \times N} | 0_{N \times (P-N)}], \mathbf{S}) = [\mathbf{S} | 0]$, $\text{Lip}(\Phi; \mathbf{X}_0) \leq \sqrt{N}$. In the following, we present three different approaches to bounding the Lipschitz constants of an mGCF that directly analyze the matrix format (11) and avoid introducing feature dimension terms.

A. Triangle Inequality Bounds

The first method is to bound the output difference of an mGCF by the triangle inequality and the submultiplicativity of matrix norms. The following proposition can be viewed as an extension of [29, Proposition 1] to the matrix case.

Proposition 6: An mGCF γ defined on a GSO domain $\mathcal{S} \subset \mathbb{R}^{N \times N}$ with parameters $\{\Theta_k\}_{k \in \mathcal{K}}$ satisfy

$$\text{Lip}(\gamma; \mathbf{S}) \leq \sum_{k \in \mathcal{K}} \bar{s}^k \|\Theta_k\|_2, \quad (15a)$$

$$\overline{\text{Lip}}(\gamma) \leq \sum_{k \in \mathcal{K}} k \bar{s}^{k-1} \|\Theta_k\|_2, \quad (15b)$$

where $\bar{s} = \sup_{\mathbf{S} \in \mathcal{S}} \|\mathbf{S}\|_2$.

Proof: Given in Appendix S3-A. ■

Proposition 6 shows that both signal and structure Lipschitz constants of an mGCF can be uniformly bounded by weighted sums of the spectral norms of the convolution coefficient matrices. The bounds scale with feature dimension implicitly through these norms, and in the case $\Theta_k = \mathbf{I}$, they are truly independent. Therefore, it is stronger than simply extending the bound in [29] by Proposition 5. Moreover, the bound does not require $\mathbf{S}$ to be symmetric and is independent of N .

Because a key relaxation in (15) is separating the perturbation on different orders of graph shift through the triangle inequality, a notable special case is when there is only one summation term ($|\mathcal{K}| = 1$), so the triangle inequality is achieved. In fact, it will be shown in Section IV-D that the bound equals the true Lipschitz constant in such a case. Therefore, we state the following bound, which is a clean instantiation of the general stability bound (7) for single-shift GCNNs.

Corollary 6.1: For a GCNN Φ with $\mathcal{K} = \{K\}$ and no layer bias, there is

$$\begin{aligned} & \|\Phi(\mathbf{X}, \mathbf{S}) - \Phi(\hat{\mathbf{X}}, \hat{\mathbf{S}})\|_{\text{F}} \\ & \leq B_{\sigma}^{L-1} \bar{s}^{KL} (\epsilon_X + \epsilon_S LK \|\mathbf{X}\|_{\text{F}} / \bar{s}) \prod_{\ell=1}^L \|\Theta_K^{(\ell)}\|_2. \end{aligned}$$

Setting both K and $\bar{s}$ to 1, Corollary 6.1 recovers Theorem 1 in [23], demonstrating the success of our bound reduction.²

B. Spectral Bounds

When multiple orders of graph shifts coexist in an mGCF, the effect of a perturbation on different orders may cancel out in (15), resulting in a loose bound. In order not to separate different orders in the analysis, we follow [25], [18] and characterize the stability of mGCFs through their frequency response. In fact, it was shown in [18, Lemma 1] that the signal Lipschitz constant of a general graph filter equals the maximum of its frequency response in absolute value, evaluated at the eigenvalues of $\mathbf{S}$. In this part, we aim to extend such analysis to structure perturbations. However, because a graph filter is defined with an eigenbasis on $\mathbf{S}$, we must show that the choice of basis does not affect the filter itself, so that any graph filter can be viewed as a function in $\mathbf{S}$ (and $\mathbf{X}$) and the Lipschitz constants (4) are well-defined.

Lemma 7: A graph filter defined in the frequency domain (2) is independent of the choice of the eigenbasis $\mathbf{V}$.

Proof: Given in Sec. S4 of the supplemental material. ■

Because an mGCF can be written as a group of GCFs summed at the output (12), we analyze the Fourier-domain counterpart with a group of graph filters aggregated in the same way, as (13). Specifically, let the function f_{pq} in (13) be a graph filter with frequency response $h_{pq}(\lambda)$, then by Lemma 7, we can choose any eigenbasis $\mathbf{V}$ of $\mathbf{S}$ and write the full matrix function τ as

$$\tau(\mathbf{X}, \mathbf{S})_{:,q} := \mathbf{V} \sum_{p=1}^P \text{diag}_i(h_{pq}(\lambda_i)) \mathbf{V}^T \mathbf{X}_{:,p}. \quad (16)$$

²Because the second line of (57) in [23] left out the $\mathbb{L}_{\sigma}^{L-1}$ term by mistake, the actual final bound should be $\mathbb{L}_{\sigma}^L (\epsilon_X + L \|\mathbf{Z}_{(0)}\|_{\text{F}} \epsilon_S) \prod_{\ell=1}^L \|\Theta_{(\ell)}\|_2$

The function τ represents a class of graph filters in matrix form that model mGCFs in the Fourier domain, but without the polynomial structure (hence more general). We define the *frequency response* of such a function as the collection of single frequency responses arranged into a matrix.

$$\mathbf{H}(\lambda) := \begin{bmatrix} h_{11}(\lambda) & \cdots & h_{1Q}(\lambda) \\ \vdots & \ddots & \vdots \\ h_{P1}(\lambda) & \cdots & h_{PQ}(\lambda) \end{bmatrix}, \quad (17)$$

From (12) we see that an mGCF γ can be defined as a function τ with a polynomial frequency response $\mathbf{H}(\lambda) = \sum_{k \in \mathcal{K}} \lambda^k \Theta_k$. To gain some insight into the new format (16) and its frequency response, with the input and output matrices in the Fourier domain denoted as $\tilde{\mathbf{Y}} = \mathbf{V}^T \tau(\mathbf{X}, \mathbf{S})$, $\tilde{\mathbf{X}} = \mathbf{V}^T \mathbf{X}$, there is

$$\tilde{\mathbf{Y}}_{:,i} = \tilde{\mathbf{X}}_{:,i} \mathbf{H}(\lambda_i). \quad (18)$$

Because each row in $\tilde{\mathbf{X}}$ or $\tilde{\mathbf{Y}}$ represents a frequency component, (18) shows that the frequency response determines the transfer at each frequency, similar to $[\tilde{\mathbf{y}}]_i = [\tilde{\mathbf{x}}]_i h(\lambda_i)$ in the single-filter case (2).

In the following, we study the stability of τ on a generic GSO domain with symmetry and bounded spectrum

$$\mathcal{S}^* := \{\mathbf{S} \in \mathbb{R}^{N \times N} : \mathbf{S} = \mathbf{S}^T, \lambda(\mathbf{S}) \subset [\underline{\lambda}, \bar{\lambda}]\}. \quad (19)$$

The set $\mathcal{S}^*$ contains all types of GSOs within the spectral range, and captures the exact conditions that graph filters and Lipschitz bounds rely on. Symmetry is assumed so that the eigendecomposition $\mathbf{S} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$ exists. The Lipschitz and integral-Lipschitz conditions on the frequency response $h(\lambda) = \sum_{k=0}^{\infty} \lambda^k \theta_k$ used in previous analyses [25], [58] hold on $\mathbb{R}$ only when the filter coefficients $\{\theta_k\}_{k \geq 1}$ have infinite support, because otherwise the frequency response is a polynomial which must diverge at $\pm\infty$. Since actual GCFs have finite order, the bounded spectra of GSOs are a key condition for the Lipschitz bounds to hold in the precise sense. With these preparations, we give an expression for the signal Lipschitz constant of graph filter groups, which generalizes [18, Lemma 1] nontrivially.

Proposition 8: The function τ (16) defined on $\mathcal{S}^*$ by a frequency response $\mathbf{H}(\lambda)$ satisfies

$$\text{Lip}(\tau; \mathbf{S}) = \max_{\lambda \in \lambda(\mathbf{S})} \|\mathbf{H}(\lambda)\|_2. \quad (20)$$

Proof: See Appendix S2-A. ■

To obtain an input-independent bound, we simply relax the set $\lambda(\mathbf{S})$ to the full spectral range $[\underline{\lambda}, \bar{\lambda}]$, so

$$\text{Lip}(\tau; \mathbf{S}) \leq \max_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}(\lambda)\|_2, \quad \forall \mathbf{S} \in \mathcal{S}^*. \quad (21)$$

This bound improves over (15a), since $\sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}(\lambda)\|_2 \leq \sup_{|\lambda| \leq \bar{s}} \sum_{k \in \mathcal{K}} |\lambda|^k \|\Theta_k\|_2 = \sum_{k \in \mathcal{K}} \bar{s}^k \|\Theta_k\|_2$, while requiring the stronger condition that the GSOs are symmetric. Strict improvement occurs only when $|\mathcal{K}| > 1$ and is more pronounced when cancellation among different graph shift terms is greater, which is likely to happen when $\mathcal{K}$ is large, or Θ_k are multiples of the same Θ with alternating signs.

A structure Lipschitz bound can also be derived using the frequency response, which improves over (15b) similarly when $\mathcal{K}$ is large, by avoiding the triangle inequality.

Proposition 9: The function τ (16) defined on $\mathcal{S}^*$ by a frequency response $\mathbf{H}(\lambda)$ satisfies

$$\begin{aligned} & \left\| \tau(\mathbf{X}, \mathbf{S}) - \tau(\mathbf{X}, \hat{\mathbf{S}}) \right\|_{\text{F}} \\ & \leq \|\mathbf{X}\|_{\text{F}} \left(\|\mathbf{S} - \hat{\mathbf{S}}\|_2 \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}'(\lambda)\|_2 + 4 \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}(\lambda)\|_2 \right) \end{aligned} \quad (22)$$

where $\mathbf{H}'(\lambda)$ is the entry-wise derivative of $\mathbf{H}(\lambda)$, and the second summation term in (22) can be left out if there is some orthogonal $\mathbf{V}$ that diagonalizes $\mathbf{S}$ and $\hat{\mathbf{S}}$ simultaneously.

Proof: Given in Appendix S2-B. ■

Proposition 9 shows that the structure Lipschitz constant is controlled by the suprema of the spectral norm of both the frequency response and its derivative. On a restricted domain with some fixed eigenbasis $\mathbf{U} \in \mathbb{R}^{N \times N}$, $\mathbf{U}\mathbf{U}^T = \mathbf{I}$, $\tilde{\mathcal{S}} = \{\mathbf{U} \text{diag}(\boldsymbol{\theta}) \mathbf{U}^T : [\boldsymbol{\theta}]_i \in [\underline{\lambda}, \bar{\lambda}]\}$, there is $\bar{\text{Lip}}(\tau|_{\tilde{\mathcal{S}}}) \leq \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}'(\lambda)\|_2$. In general, the eigenbasis misalignment term makes the bound linear in $\|\mathbf{X}\|_{\text{F}}$ and affine in $\|\mathbf{S} - \hat{\mathbf{S}}\|_2$. Theorem 1 can handle the affine bound as well, by adding the term $4 \sup \|\mathbf{H}'_{\ell}(\lambda)\|_2$ to $C_{\ell} \epsilon_{\mathcal{S}}$. Because the structure Lipschitz bound in (15b) can be viewed as a relaxation of $\sup \|\mathbf{H}'(\lambda)\|_2$ similarly, the bound (22) can make an improvement when the additional intercept term is offset by the looseness of the triangle inequality.

Because Propositions 8 and 9 do not assume polynomial frequency responses, they apply to GNNs and graph filters defined with other graph filters in place of the GCF in (12), such as PPNP [49], ARMA_K [59] and CayleyNet [60] with rational filters and GraphHeat with heat diffusion kernels.

C. Structure Lipschitz Bound with Frobenius Norm

The difficulty of deriving a strict structure Lipschitz bound for τ can be circumvented by using the Frobenius norm for the GSO difference. The following proposition expresses the exact normalized Lipschitz constant of τ on $\mathcal{S}^*$, which can be viewed as a counterpart of Proposition 8, albeit with a different norm.

Proposition 10: Let the function τ (16) be defined on $\mathcal{S}^*$ by a frequency response $\mathbf{H}(\lambda)$, and $\bar{\text{Lip}}_{\text{F}}$ be the uniform structure Lipschitz constant (5) using the Frobenius norm instead of the spectral norm for measuring $\mathbf{S} - \hat{\mathbf{S}}$ in (4). We have

$$\bar{\text{Lip}}(\tau)/\sqrt{N} \leq \bar{\text{Lip}}_{\text{F}}(\tau) = \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}'(\lambda)\|_2. \quad (23)$$

Proof: See Appendix S2-C. ■

The bound on $\bar{\text{Lip}}(\gamma)$ is obtained by relaxing the Frobenius norm of the difference to the spectral norm, at the cost of $\sqrt{N}$. With the spectral norm distance, (23) combined with (20) and (7) yields a GCNN structure stability bound that is no worse than the one in [25] under a common set of conditions.

Proposition 11: Compared to combining Propositions 8 and 10 and Theorem 1, the bound in [25] with absolute perturbation cannot be smaller and can be $\prod_{\ell=1}^L \sqrt{F_{\ell} F_{\ell+1}}$ times as large.

Proof: See Sec. S5 of the supplemental material. ■

The improvement shown in Proposition 11 originates from two parts. First, we characterize the stability of mGCFs directly, so there isn't a $\sqrt{PQ}$ factor that would appear if GCF bounds are used. Because $\|\mathbf{A}\|_2 \leq \sqrt{PQ} \|\mathbf{A}\|_{\text{max}}$, the increase in norm is always canceled out by the removal of the dimension terms. In the best case, the two norms coincide (e.g., for scaled identity matrices), so the bound via mGCF can be tighter. Secondly, we bypass the eigenvector misalignment term $\delta = (\|\mathbf{U} - \mathbf{V}\|_2 + 1)^2 - 1$ between $\mathbf{S}$ and $\mathbf{E}$: even when $\hat{\mathbf{S}}$ is restricted in a small spectral-norm ball around $\mathbf{S}$, the difference $\mathbf{E}$ can have an arbitrary eigenbasis, and δ is only bounded by 8.

D. Lower Bound and Tightness

Now we discuss the tightness of the proposed upper bounds. Proposition 8 gives an exact expression for the signal Lipschitz constant of any mGCF. The relaxed input-independent bound converges to the true value when $N \rightarrow \infty$ and the spectrum of $\mathbf{S}$ uniformly spans the whole range $[\underline{\lambda}, \bar{\lambda}]$. For structure Lipschitz constants of the mGCF, the bounds given in Proposition 6, 9 and 10 all reduce to $\bar{\text{Lip}}(\gamma) \sim \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}'(\lambda)\|_2$, under $|\mathcal{K}| = 1$, $N = 1$ and $\mathbf{H}(\lambda) = 0$, respectively. In the following proposition, we show that it is indeed a lower bound.

Proposition 12: Let γ be an mGCF defined on $\mathcal{S}^*$, then

$$\begin{aligned} \text{Lip}(\gamma; \mathbf{X}) & \geq \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{X} \mathbf{H}'(\lambda)\|_{\text{F}}, \\ \bar{\text{Lip}}(\gamma) & \geq \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}'(\lambda)\|_2, \end{aligned} \quad (24)$$

and both strict inequality holds for some $\underline{\lambda}, \bar{\lambda}, \{\boldsymbol{\Theta}_k\}$ and $N > 1$.

Proof: Given in Appendix S3-B. ■

Proposition 12 shows that our upper bounds are near-optimal. Firstly, for mGCFs with $|\mathcal{K}| = 1$ and symmetric GSOs, the bound (15b) gives the exact structure Lipschitz constant, and generally the slackness depends on how centralized the energy is across different graph shift orders. Secondly, because the bound (23) cannot hold with a constant less than one, it is tight up to a constant for any fixed N . Lastly, when $\|\mathbf{H}(\lambda)\| / \|\mathbf{H}'(\lambda)\| \rightarrow 0$ or $\epsilon_{\mathcal{S}} \rightarrow \infty$, the bound (22) also matches the lower bound. We also remark that this lower bound can be larger than considering only the finite set of a certain type of GSOs of N nodes, because $\mathcal{S}^*$ is a surrogate for actual GSO domains where $\mathbf{S}$ and $\hat{\mathbf{S}}$ can take continuous values that may not correspond to a GSO.

Moreover, Proposition 12 shows that the *instability* of an mGCF grows with the variation rate of its frequency response, which constitutes a discriminability-stability tradeoff under GSO perturbations. Such a tradeoff cannot be demonstrated by upper bounds, because they only show the *stability* of a filter.

V. DISCUSSIONS

A. Difficulty of Tighter Local Lipschitz Bounds

Our upper bounds on the structure Lipschitz constants of mGCFs (Proposition 6, 10, and 9) are inherently global, i.e.,

independent of the inputs $\mathbf{X}, \hat{\mathbf{X}}, \mathbf{S}$ and $\hat{\mathbf{S}}$. This is because when N becomes sufficiently large, the spectrum of $\mathbf{S}$ typically spans the entire interval $[\underline{\lambda}, \bar{\lambda}]$ evenly, and there is no significant gain in tightness by stating the bounds with $\lambda(\mathbf{S})$. Although Theorem 1 can be adapted for changes local to $\mathbf{S}$, there seems to be no concrete local bound that meaningfully improves over global ones. This might be explained by the fact that in $\mathcal{S}^*$, the eigenspace misalignment between $\mathbf{S}$ and $\hat{\mathbf{S}}$ (related to Proposition 9) or between $\mathbf{S}$ and $\mathbf{E}$ (required by bounds in [25]) cannot be effectively bounded. However, restricting $\mathbf{S}$ to the discrete domain of a certain GSO poses new challenges to the analysis.

B. Implications of Finite Spectrum

Table I shows that for most GCNN models, the GSO is normalized and has a spectral radius bounded by some constant. Even for unnormalized GSOs, the spectral radius has an upper bound depending on N , which is fixed in our setting. The structure Lipschitz constant can be bounded by the supremum of $\|\mathbf{H}'(\lambda)\|_2$, which equals the Lipschitz constant of $\mathbf{H}(\lambda)$ with respect to the spectral norm. Our observation is vastly different from [25], [58] by accounting for bounded spectra: The relative GSO distance aimed at preventing bound blow-up when $\|\mathbf{S}\|_2 \rightarrow \infty$ is relevant only for GCNNs using certain unnormalized GSOs, such as the Laplacian for connected graphs. Although enabling free variations of the frequency response near $\lambda \approx 0$, the integral Lipschitz condition actually poses a stricter condition when $\lambda > 1$. To make things worse, the asymptotic bounds derived under relative distance can be less accurate for large λ . In contrast, we show that the structure stability of a GCNN is determined by variations in the frequency response of the mGCFs uniformly over the spectral range, and its ability to distinguish different frequency components need not rely on nonlinearity. Moreover, since the stability bounds scale with λ^K , we explain why most practically successful GCNNs are defined using GSOs with spectral range $[-1, 1]$, or only contain first-order convolutions.

VI. NUMERICAL EXPERIMENTS

We numerically validated and compared the proposed stability bounds using both synthetic graph filters and real-world GCNNs. Because structural stability is the more difficult problem, the comparison of signal stability bounds is omitted for brevity.

A. Synthetic Graph Filters

The structure Lipschitz bounds for mGCFs are compared using synthetic heat-diffusion kernels and input signals. In this experiment, all entries of a randomly generated matrix are sampled independently. The frequency response of a heat kernel is given by $h(\lambda) = e^{-\tau\lambda}$. We construct an mGCF with weights $[\Theta_k]_{pq} = \frac{\alpha(-\tau)^k}{k!}$, $\alpha \sim \mathcal{U}[0, 1]$, $\tau \sim \mathcal{U}[0, 10]$, so that each component GCF is a Taylor series expansion of a heat kernel with i.i.d. decay and scaling parameters. The number of nodes and signals is fixed at $N = 100$ and $F = 16$, and $[\mathbf{X}]_{ip} \sim \mathcal{N}(0, 1)$. The GSO is the symmetrically

normalized Laplacian, with each pair of nodes having an edge with probability 0.2. Three types of GSO perturbations are considered: *Random edge flips* randomly choose a number of edges and flip them, *Gaussian noise* samples $[\mathbf{E}]_{ij} \sim \mathcal{N}(0, 1/N)$ and averages it with its transpose, and *Scaled identity*, where $\mathbf{E} = \epsilon \mathbf{I}$. The number of edges for the first and the scaling factors of the last two perturbations are sampled from exponential distributions. Moreover, the last two methods clamp the eigenvalues of $\hat{\mathbf{S}}$ into $[0, 2]$.

Results for different choices of $\mathcal{K}$ and F are plotted in Figure 1. The bounds are generally looser as the filter order increases, with the triangle inequality bound nearly achieved when $\mathcal{K} = \{1\}, F = 1$. In all cases, *scaled identity* produces the strongest output perturbations, and *Gaussian noise* produces the same or slightly stronger perturbations than *Random edge flips*. Among the bounds, the triangle-inequality bound is the tightest strict structure Lipschitz bound when $\max \mathcal{K}$ or $\|\mathbf{E}\|_2$ is small, and is surpassed by the spectral bound otherwise. The looseness of the triangle inequality is demonstrated by the gap between the triangle and ∞ -asymptotic bounds, which increased drastically at large K , rendering the triangle inequality bound vacuous. The proposed asymptotic bound at 0 consistently outperforms the baseline, while being relatively loose for small K , due to the $\sqrt{N}$ factor. The lower bound closely matches the upper bounds in all scenarios, and is attained by the scaled identity perturbation for $K = 1$. Because $\mathbf{S}$ is fixed, the perturbation samples are far from reaching the lower bound for higher K , because the lower bound is taken globally on $\mathcal{S}^*$.

B. Practical GCNNs

Next, we tested the GCNN stability bound (7) on four GCNN models: GCN, SGC, ChebNet, and GIN. The first three models were trained on the dataset Cora [61] for semi-supervised node classification, which has $N = 2708$ nodes. GIN was trained on the PROTEINS dataset [62] for graph classification, which contains 1113 graphs with $4 \sim 620$ nodes. The order of graph convolution for each layer was set to $\mathcal{K} = \{0, 1, 2\}$ in ChebNet and to $\mathcal{K} = \{10\}$ in SGC. The GIN has an input dropout of $p = 0.5$ for each layer. GCN, SGC, and ChebNet have feature dimension $F_\ell = 64$ for all hidden layers, while GIN has $F_\ell = 16$. All models are trained with the Adam optimizer with a learning rate of 0.01 and a weight decay of 5×10^{-4} for 200 epochs. A stability analysis for the trained models is presented in Table II, where the GSO partial Lipschitz constant of the network is given as an affine function of $x = \|\mathbf{X}\|_F$. The signal and structure Lipschitz bounds of each layer were computed using the spectral bound (21) and the triangle inequality bound (15b), respectively.

For the first-order GCNNs (GCN and GIN), we plot the stability bound using only the triangle-inequality structure Lipschitz bound for layers, along with random edge flips, since other bounds cannot be better in such a case. To obtain a realistic high- K model, a ChebNet with $K = 7$ is trained on the denser heterophilous Squirrel dataset (filtered) [63] with the same hyperparameters, achieving a test accuracy of 35.2%. For GCN and ChebNet, the scaled identity perturbation and a

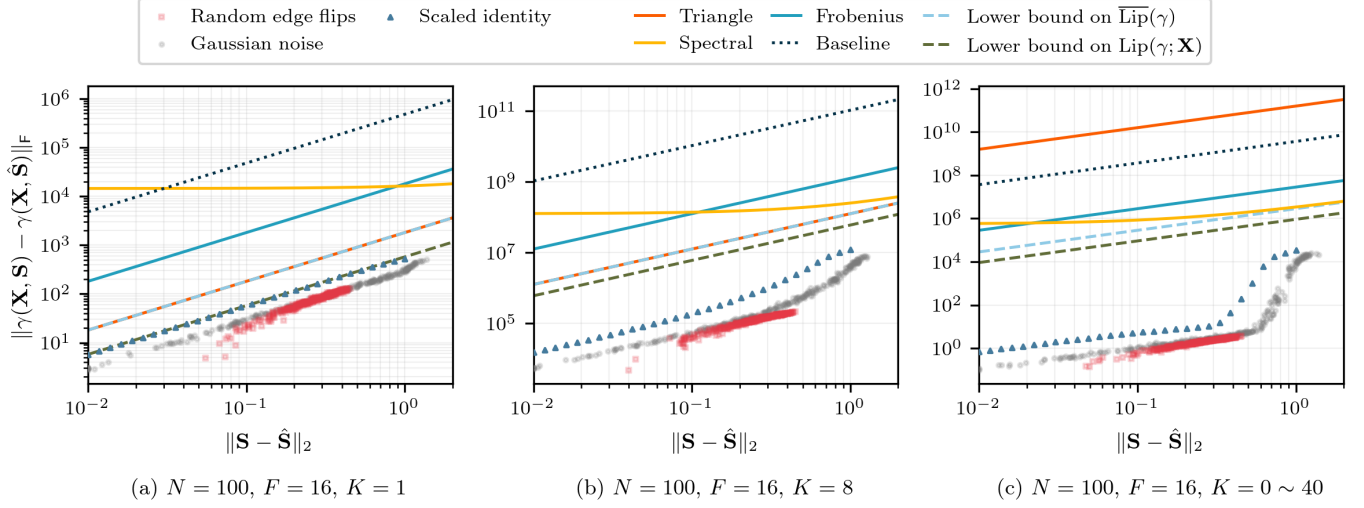


Fig. 1: Comparisons between bounds and various perturbations on synthetic data. Each line style represents a different type of bound. Reference for formulae: ‘Triangle’–(15b), ‘Frobenius’–(23), ‘Lower bounds’–(24), ‘Spectral’–(22), ‘Baseline’–(33). The triangle-inequality bound matches the lower bound for $\text{Lip}(\gamma)$ when $|K| = 1$, so they overlap with each other.

Model	ℓ	B_ℓ	C_ℓ	E_ℓ	$\text{Lip}(\Phi; \mathbf{X})$
GCN	1	4.53	4.53	66.1	$38.3x + 280$
	2	4.23	4.23	24.2	
GCN (no bias)	1	4.58	4.58	0	$38.8x$
	2	4.24	4.24	0	
SGC	1	11.9	119	102	$119x$
ChebNet	1	3.13	5.96	33.9	$41.0x + 227$
	2	3.36	6.70	5.05	
GIN	1	36.1	1.40	6.02	$67.9x + 8.96$ ($N = 26$)
	2	0.980	0	3.41	
	3	23.2	0.893	4.23	
	4	1.08	0	2.55	

TABLE II: Stability analysis for the four GCNN instances. The Lipschitz constants and the bias terms ($E_\ell = \sqrt{N} \|\mathbf{b}^{(\ell)}\|_2$) are displayed for each layer. The N used to calculate the stability bound for GIN is set to the dataset’s median. The input signal magnitude for Cora: $\|\mathbf{X}\|_F = 221.8$

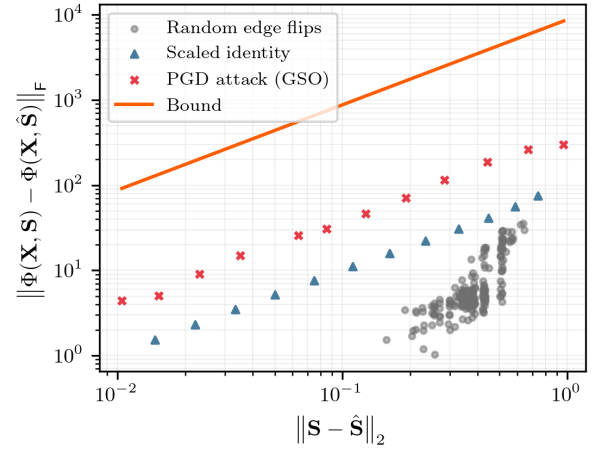


Fig. 2: Plot of the triangle-inequality bound and perturbations for GCN. Other bounds are excluded because they are suboptimal in the linear case. The PGD attack was run 3 times for each perturbation budget.

projected gradient descent (PGD) attack on the GSO domain based on [64] were also plotted (pseudocode is given in Section S6 of the supplemental material). For GIN, the bound (7) is calculated for every randomly chosen graph sample in the whole dataset using the sample’s N and $\mathbf{X}$, and $\bar{s}$ is set to $\max\{\|\mathbf{S}\|_2, \|\hat{\mathbf{S}}\|_2\}$. The perturbation magnitudes are plotted as fractions of the bound.

The results are given in Figures 2, 3, and 4. A general trend is that the bounds become looser as the number of layers or the order of graph convolution increases, which matches the values in Table II. The PGD attack achieved significantly larger output deviations than edge flips and acts as an empirical lower bound on the instability of a GCNN. At higher K , the spectral bound becomes much tighter than the triangle-inequality bound, which is consistent with the comparison of the bounds for a single mGCF.

VII. CONCLUSION

In this work, we derived upper bounds of the Lipschitz constants of GCNNs with respect to signal and structure perturbations to the inputs. Because the bounds are finite regardless of the coefficients of the underlying GCFs, there is no clear distinction between stable and unstable GCNNs, while those with smaller Lipschitz constants are more stable. We characterized the spectral properties of the filters in each layer of a GCNN as a matrix function and derived order-matching upper and lower bounds on the Lipschitz constants under finite spectrum ranges, which significantly improve over existing results. We showed that the stability of a GCNN does not always degrade as the dimensions of the hidden layers increase, as long as the spectral norms of the coefficient matrices are controlled. Overall, the novel stability bounds

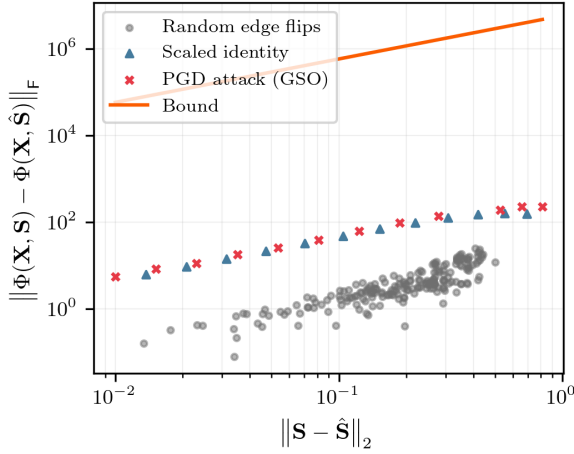


Fig. 3: Comparison of perturbations and different bounds for a ChebNet with $K = 7$. The bounds asymptotic to 0 were excluded, as the relatively large N renders them void.

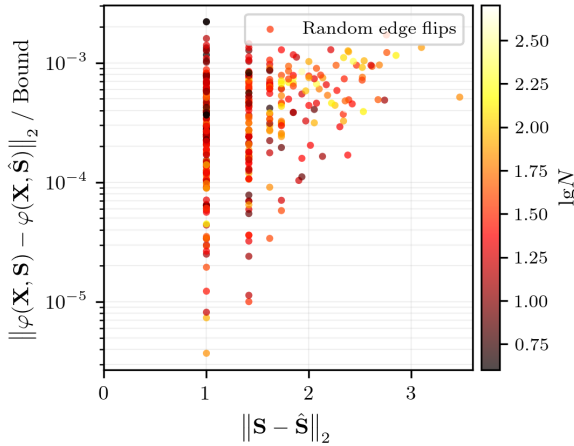


Fig. 4: Comparison between random edge perturbations and per-graph stability bound for GIN. The y-axis shows the proportion of output deviation to the bound. The color of the markers indicates the number of nodes in that sample.

help understand the vulnerability of GCNNs and provide valuable insights into designing more robust GCNNs.

APPENDIX A

ON THE VALIDITY OF THE RELATIVE GSO DISTANCE

The relative GSO distance is defined as $d_R(\mathbf{S}, \hat{\mathbf{S}}) = \inf_{\mathbf{E} \in \Delta(\mathbf{S}, \hat{\mathbf{S}})} \|\mathbf{E}\|_2$, where

$$\Delta(\mathbf{S}, \hat{\mathbf{S}}) = \{\mathbf{E} \in \mathbb{R}^{N \times N} : \hat{\mathbf{S}} = \mathbf{S} + \mathbf{E}\mathbf{S} + \mathbf{S}\mathbf{E}\}.$$

Let the eigendecomposition of $\mathbf{S}$ be $\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, then the difference in GSO expressed in the spectral domain can be written as

$$\begin{aligned} \tilde{\mathbf{D}} &:= \mathbf{V}^T(\mathbf{E}\mathbf{S} + \mathbf{S}\mathbf{E})\mathbf{V} \\ &= \mathbf{V}^T(\mathbf{E}\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T + \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T\mathbf{E})\mathbf{V} \\ &= \mathbf{V}^T\mathbf{E}\mathbf{V}\mathbf{\Lambda} + \mathbf{\Lambda}\mathbf{V}^T\mathbf{E}\mathbf{V}, \end{aligned}$$

therefore $[\tilde{\mathbf{D}}]_{ij} = (\lambda_i + \lambda_j)[\mathbf{V}^T\mathbf{E}\mathbf{V}]_{ij}$. Because this term is zero if $\lambda_i + \lambda_j = 0$ and can take any value otherwise, since $\mathbf{V}$ is invertible, $\Delta(\mathbf{S}, \hat{\mathbf{S}})$ is nonempty (so that d_R is well-defined) iff

$$\hat{\mathbf{S}} - \mathbf{S} \in \{\mathbf{V}\tilde{\mathbf{D}}\mathbf{V}^T : \lambda_i + \lambda_j = 0 \Rightarrow [\tilde{\mathbf{D}}]_{ij} = 0\}.$$

When we set

$$\mathcal{S} = \{\text{diag}(\mathbf{A}\mathbf{1}) - \mathbf{A} : \mathbf{A} \in \{0, 1\}^{N \times N} \text{ connected}\},$$

let $\mathbf{L}$ be a Laplacian with eigendecomposition $\mathbf{L} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, then $\lambda_1 \geq \dots \geq \lambda_{N-1} > \lambda_N = 0$ and $\mathbf{v}_N = \mathbf{1}$ [65]. We have $\lambda_i + \lambda_j = 0$ iff $i = j = N$ and $\mathbf{V}^T(\tilde{\mathbf{L}} - \mathbf{L})\mathbf{V} = \mathbf{V}^T[*|0] - \mathbf{\Lambda}$, whose (N, N) -element is zero, so any such difference can be represented by some $\mathbf{E}$.

However, if $\mathbf{S} = 0$ is the adjacency matrix of a graph with N nodes and no edges, $\hat{\mathbf{S}}$ must also be 0, so d_R fails to measure all pairs of adjacency matrices. Also, let $\mathbf{L}$ be the graph Laplacian of the undirected graph with 4 nodes and edges $(1, 2), (3, 4)$, and let $\tilde{\mathbf{L}}$ be the perturbed Laplacian obtained by adding the edge $(2, 3)$. Calculation shows that no $\mathbf{E}$ can represent this difference.

In fact, when the spectral range is bounded by some constant, the absolute perturbation is no larger than the relative perturbation by a constant factor (there is $\|\mathbf{E}\|_2 = \|\hat{\mathbf{E}}\mathbf{S} + \mathbf{S}\hat{\mathbf{E}}\|_2 \leq 2\bar{\rho}\|\hat{\mathbf{E}}\|_2$), and it is unnecessary to perform analysis using relative perturbations.

REFERENCES

- [1] G. Leus, A. G. Marques, J. M. Moura, A. Ortega, and D. I. Shuman, "Graph Signal Process.: History, development, impact, and outlook," *IEEE Signal Process. Mag.*, vol. 40, pp. 49–60, June 2023.
- [2] A. Sandryhaila and J. M. F. Moura, "Discrete Signal Process. on Graphs," *IEEE Trans. Signal Process.*, vol. 61, pp. 1644–1656, Apr. 2013.
- [3] L. Ruiz, F. Gama, and A. Ribeiro, "Graph Neural Networks: Architectures, Stability, and Transferability," *Proc. IEEE*, vol. 109, pp. 660–682, May 2021.
- [4] E. Isufi, F. Gama, D. I. Shuman, and S. Segarra, "Graph filters for signal processing and machine learning on graphs," *IEEE Trans. Signal Process.*, vol. 72, pp. 4745–4781, 2024.
- [5] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, "Neural message passing for quantum chemistry," in *Int. Conf. Mach. Learn.*, pp. 1263–1272, Pmlr, 2017.
- [6] Y. Shen, J. Zhang, S. H. Song, and K. B. Letaief, "Graph neural networks for wireless communications: From theory to practice," *IEEE Trans. Wireless Commun.*, vol. 22, no. 5, pp. 3554–3569, 2023.
- [7] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, and M. Wang, "LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation," July 2020.
- [8] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," *arXiv preprint arXiv:1609.02907*, 2016.
- [9] E. Ceci and S. Barbarossa, "Graph Signal Process. in the Presence of Topology Uncertainties," *IEEE Trans. Signal Process.*, vol. 68, pp. 1558–1573, 2020.
- [10] S. Segarra, A. G. Marques, G. Mateos, and A. Ribeiro, "Network topology inference from spectral templates," *IEEE Trans. Signal Inf. Process. Netw.*, vol. 3, no. 3, pp. 467–483, 2017.
- [11] A. Fascista, A. Coluccia, and C. Ravazzi, "Graph signal reconstruction under heterogeneous noise via adaptive uncertainty-aware sampling and soft classification," *IEEE Trans. Signal Inf. Process. Netw.*, vol. 10, pp. 277–293, 2024.
- [12] K. Xu, H. Chen, S. Liu, P.-Y. Chen, T.-W. Weng, M. Hong, and X. Lin, "Topology attack and defense for graph neural networks: an optimization perspective," in *Proc. 28th Int. Joint Conf. Artif. Intell.*, p. 3961–3967, 2019.

- [13] D. Zügner and S. Günnemann, “Certifiable robustness and robust training for graph convolutional networks,” in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, (New York, NY, USA), p. 246–256, Association for Computing Machinery, 2019.
- [14] E. Isufi, A. Loukas, A. Simonetto, and G. Leus, “Filtering Random Graph Processes Over Random Time-Varying Graphs,” *IEEE Trans. Signal Process.*, vol. 65, pp. 4406–4421, Aug. 2017.
- [15] M. Eisen and A. Ribeiro, “Optimal Wireless Resource Allocation With Random Edge Graph Neural Networks,” *IEEE Trans. Signal Process.*, vol. 68, pp. 2977–2991, 2020.
- [16] S. Geisler, T. Schmidt, H. Şirin, D. Zügner, A. Bojchevski, and S. Günnemann, “Robustness of graph neural networks at scale,” in *Proc. 35th Int. Conf. Neural Inf. Process. Syst.*, (Red Hook, NY, USA), 2021.
- [17] H. Wu, C. Wang, Y. Tyshetskiy, A. Docherty, K. Lu, and L. Zhu, “Adversarial examples for graph data: deep insights into attack and defense,” in *Proc. 28th Int. Joint Conf. Artif. Intell.*, p. 4816–4823, 2019.
- [18] R. Arghal, E. Lei, and S. S. Bidokhti, “Robust Graph Neural Networks via Probabilistic Lipschitz Constraints,” in *4th Annual Conf. Learn. for Dynamics and Control*, 2022.
- [19] Y. Jia, D. Zou, H. Wang, and H. Jin, “Enhancing Node-Level Adversarial Defenses by Lipschitz Regularization of Graph Neural Networks,” in *Proc. 29th ACM SIGKDD Conf. Knowledge Discovery and Data Mining*, (Long Beach CA USA), pp. 951–963, Aug. 2023.
- [20] S. I. Juvina, A. A. Neacşu, J. Rony, J.-C. Pesquet, C. Burileanu, and I. B. Ayed, “Training graph neural networks subject to a tight lipschitz constraint,” *Trans. Mach. Learn. Research*, 2025.
- [21] Z. Gao and E. Isufi, “Learning Stable Graph Neural Networks via Spectral Regularization,” in *2022 56th Asilomar Conf. Signals, Syst., and Computers*, pp. 01–05, Oct. 2022.
- [22] Y. Tsuzuku, I. Sato, and M. Sugiyama, “Lipschitz-margin training: scalable certification of perturbation invariance for deep neural networks,” in *Proc. 32nd Int. Conf. Neural Inf. Process. Syst.*, (Red Hook, NY, USA), p. 6542–6551, 2018.
- [23] V. K. Singh, S. Kumar, A. Prasad, and Jayadeva, “A Unified Optimization-Based Framework for Certifiably Robust and Fair Graph Neural Networks,” *IEEE Trans. Signal Process.*, vol. 73, pp. 83–98, 2025.
- [24] H. Kenlay, D. Thanou, and X. Dong, “On the stability of graph convolutional neural networks under edge rewiring,” in *Int. Conf. Acoust., Speech and Signal Process.*, pp. 8513–8517, 2021.
- [25] F. Gama, J. Bruna, and A. Ribeiro, “Stability Properties of Graph Neural Networks,” *IEEE Trans. Signal Process.*, vol. 68, pp. 5680–5695, 2020.
- [26] A. Parada-Mayorga and A. Ribeiro, “Algebraic Neural Networks: Stability to Deformations,” *IEEE Trans. Signal Process.*, vol. 69, pp. 3351–3366, 2021.
- [27] L. Testa, C. Battiloro, S. Sardellitti, and S. Barbarossa, “Stability of Graph Convolutional Neural Networks Through The Lens of Small Perturbation Analysis,” in *IEEE Int. Conf. Acoustics, Speech and Signal Process.*, pp. 6865–6869, Apr. 2024.
- [28] H. Kenlay, D. Thanou, and X. Dong, “On the stability of polynomial spectral graph filters,” in *IEEE Int. Conf. Acoustics, Speech and Signal Process.*, (Barcelona, Spain), pp. 5350–5354, May 2020.
- [29] H. Kenlay, D. Thanou, and X. Dong, “Interpretable Stability Bounds for Spectral Graph Filters,” in *Proc. 38th Int. Conf. Mach. Learn.*, pp. 5388–5397, July 2021.
- [30] L. Ruiz, Z. Wang, and A. Ribeiro, “Graphon and Graph Neural Network Stability,” in *IEEE Int. Conf. Acoustics, Speech and Signal Process.*, pp. 5255–5259, June 2021.
- [31] Z. Wang, L. Ruiz, and A. Ribeiro, “Stability to deformations of manifold filters and manifold neural networks,” *IEEE Trans. Signal Process.*, vol. 72, pp. 2130–2146, 2024.
- [32] V. A. Martirosyan, D. Malatesta, H. Talbot, J. H. Giraldo, and F. D. Malliaros, “Generalization bounds for spectral GNNs via fourier domain analysis,” in *The 29th Int. Conf. Artif. Intell. Statist.*, 2026.
- [33] A. Parada-Mayorga, Z. Wang, F. Gama, and A. Ribeiro, “Stability of Aggregation Graph Neural Networks,” *IEEE Trans. Signal Inf. Process. Netw.*, vol. 9, pp. 850–864, 2023.
- [34] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, “Graph Attention Networks,” Feb. 2018.
- [35] W. L. Hamilton, R. Ying, and J. Leskovec, “Inductive Representation Learning on Large Graphs,” Sept. 2018.
- [36] V. Dwivedi and X. Bresson, “A generalization of transformer networks to graphs,” in *AAAI Workshop on Deep Learning on Graphs: Methods and Applications*, 2021.
- [37] Y. Huang, W. Lu, J. Robinson, Y. Yang, M. Zhang, S. Jegelka, and P. Li, “On the stability of expressive positional encodings for graphs,” in *The 12th Int. Conf. Learn. Representations*, 2024.
- [38] G. Dasoulas, K. Scaman, and A. Virmaux, “Lipschitz normalization for self-attention layers with application to graph neural networks,” in *Proc. 38th Int. Conf. Mach. Learn.*, vol. 139, pp. 2456–2466, 2021.
- [39] Z. Gao, E. Isufi, and A. Ribeiro, “Stability of graph convolutional neural networks to stochastic perturbations,” *Signal Process.*, vol. 188, p. 108216, 2021.
- [40] L. Ruiz, L. F. O. Chamon, and A. Ribeiro, “Transferability properties of graph neural networks,” *IEEE Trans. Signal Process.*, vol. 71, pp. 3474–3489, 2023.
- [41] X. Wang, E. Ollila, and S. A. Vorobyov, “Graph convolutional neural networks sensitivity under probabilistic error model,” *IEEE Transactions on Signal and Information Processing over Networks*, vol. 10, pp. 788–803, 2024.
- [42] Y. Abbahaddou, S. Ennadir, J. F. Lutzeyer, M. Vazirgiannis, and H. Boström, “Bounding the expected robustness of graph neural networks subject to node feature attacks,” in *The 12th Int. Conf. Learn. Representations*, 2024.
- [43] N. Zhang, H. Kenlay, L. Zhang, M. Cucuringu, and X. Dong, “On the stability of graph convolutional neural networks: A probabilistic perspective,” in *The Thirty-ninth Annual Conf. Neural Inf. Process. Syst.*, 2025.
- [44] M. Defferrard, X. Bresson, and P. Vandergheynst, “Convolutional neural networks on graphs with fast localized spectral filtering,” in *Advances in Neural Inf. Process. Syst.*, (Red Hook, NY, USA), p. 3844–3852, 2016.
- [45] F. Wu, T. Zhang, A. H. d. Souza, C. Fifty, T. Yu, and K. Q. Weinberger, “Simplifying Graph Convolutional Networks,” June 2019.
- [46] B. Xu, H. Shen, Q. Cao, K. Cen, and X. Cheng, “Graph convolutional networks using heat kernel for semi-supervised learning,” in *Proc. 28th Int. Joint Conf. Artif. Intell.*, p. 1928–1934, 2019.
- [47] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, “How powerful are graph neural networks?,” in *Int. Conf. Learn. Representations*, 2019.
- [48] M. He, Z. Wei, and J.-R. Wen, “Convolutional neural networks on graphs with chebyshev approximation, revisited,” in *Advances in Neural Inf. Process. Syst.*, vol. 35, pp. 7264–7276, 2022.
- [49] J. Gasteiger, A. Bojchevski, and S. Günnemann, “Combining neural networks with personalized pagerank for classification on graphs,” in *Int. Conf. Learn. Representations*, 2019.
- [50] M. O. Searcoid, *Metric Spaces*. Springer Undergraduate Mathematics Series, Springer London, 2006.
- [51] F. Gama, A. Ribeiro, and J. Bruna, “Stability of graph scattering transforms,” in *Advances in Neural Information Processing Systems*, vol. 32, pp. 8038–8048, Curran Associates, Inc., 2019.
- [52] F. Gama, A. Ribeiro, and J. Bruna, “Stability of graph neural networks to relative perturbations,” in *IEEE Int. Conf. Acoust., Speech and Signal Process.*, pp. 9070–9074, 2020.
- [53] F. Serratos, “Graph edit distance: Restrictions to be a metric,” *Pattern Recognition*, vol. 90, pp. 250–256, 2019.
- [54] X. Gao, B. Xiao, D. Tao, and X. Li, “A survey of graph edit distance,” *Pattern Anal. Appl.*, vol. 13, pp. 113–129, 02 2010.
- [55] C.-Y. Chuang and S. Jegelka, “Tree Mover’s Distance: Bridging Graph Metrics and Stability of Graph Neural Networks,” in *36th Conf. Neural Inf. Process. Syst. (NeurIPS 2022)*, 2022.
- [56] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and S. Y. Philip, “A comprehensive survey on graph neural networks,” *IEEE Trans. Neural Networks and Learn. Syst.*, vol. 32, no. 1, pp. 4–24, 2020.
- [57] F. Gama, A. G. Marques, A. Ribeiro, and G. Leus, “MIMO graph filters for convolutional neural networks,” in *IEEE 19th Int. Workshop Signal Process. Advances Wireless Commun.*, 2018.
- [58] S. Pfommer, A. Ribeiro, and F. Gama, “Discriminability of single-layer graph neural networks,” in *IEEE Int. Conf. on Acoust., Speech and Signal Proc.*, pp. 8508–8512, 2021.
- [59] F. M. Bianchi, D. Grattarola, L. Livi, and C. Alippi, “Graph neural networks with convolutional arma filters,” *IEEE Trans. on Pattern Anal. and Mach. Intell.*, vol. 44, no. 7, pp. 3496–3507, 2022.
- [60] R. Levie, F. Monti, X. Bresson, and M. M. Bronstein, “CayleyNets: Graph Convolutional Neural Networks With Complex Rational Spectral Filters,” *IEEE Trans. Signal Process.*, vol. 67, pp. 97–109, Jan. 2019.
- [61] P. Sen, G. Namata, M. Bilgic, L. Getoor, B. Gallagher, and T. Eliassi-Rad, “Collective classification in network data,” *AI Mag.*, vol. 29, p. 93–106, Sept. 2008.
- [62] C. Morris, N. M. Kriege, F. Bause, K. Kersting, P. Mutzel, and M. Neumann, “Tudataset: A collection of benchmark datasets for learning with graphs,” in *Graph Representation Learning and Beyond (GRL+), ICML Workshop*, 2020.

- [63] O. Platonov, D. Kuznedelev, M. Diskin, A. Babenko, and L. Prokhorenkova, "A critical look at the evaluation of GNNs under heterophily: Are we really making progress?," in *The 11th Int. Conf. Learn. Representations*, 2023.
- [64] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Int. Conf. Learn. Representations*, 2018.
- [65] A. Brouwer and W. Haemers, *Spectra of Graphs*. Universitext, Springer New York, 2011.
- [66] J. Zhang and Z. Zhao, "Improved Stability Bounds for Graph Convolutional Neural Networks Under Graph Perturbations," in *2024 IEEE Inf. Theory Workshop (ITW)*, pp. 307–312, Nov. 2024.
- [67] G. H. Golub and C. F. Van Loan, *Matrix Computations - 4th Edition*. Philadelphia, PA: Johns Hopkins University Press, 2013.
- [68] J. Liu, C. Zhu, Z. Long, and Y. Liu, "Tensor regression," *Foundations and Trends in Mach. Learn.*, vol. 14, pp. 379–565, 2023.

Supplemental Material for “Sharp Stability Bounds for Graph Convolutional Neural Networks”

S1 PROOFS OF REDUCTIONS

A. Proof of Theorem 1

For any function $f : \mathbb{R}^{N \times P} \times \mathcal{S} \rightarrow \mathbb{R}^{N \times Q}$, the triangle inequality can separate the effect of perturbations on $\mathbf{X}$ and $\mathbf{S}$:

$$\begin{aligned} & \|f(\mathbf{X}, \mathbf{S}) - f(\hat{\mathbf{X}}, \hat{\mathbf{S}})\|_{\text{F}} \\ & \leq \|f(\mathbf{X}, \mathbf{S}) - f(\mathbf{X}, \hat{\mathbf{S}})\|_{\text{F}} + \|f(\mathbf{X}, \hat{\mathbf{S}}) - f(\hat{\mathbf{X}}, \hat{\mathbf{S}})\|_{\text{F}} \quad (25) \\ & \leq \text{Lip}(f; \mathbf{X}) \|\mathbf{S} - \hat{\mathbf{S}}\|_2 + \text{Lip}(f; \hat{\mathbf{S}}) \|\mathbf{X} - \hat{\mathbf{X}}\|_{\text{F}}, \end{aligned}$$

for any $\mathbf{S}, \hat{\mathbf{S}} \in \mathcal{S}; \mathbf{X}, \hat{\mathbf{X}} \in \mathbb{R}^{N \times F_0}$.

By (3a), when $\mathbf{S}$ is fixed, Φ is a function composition of maps between signals. It is easy to see that $\text{Lip}(\sigma) \leq \text{Lip}(\sigma) \leq B_\sigma$. By $\text{Lip}(f \circ g) \leq \text{Lip}(f)\text{Lip}(g)$,

$$\begin{aligned} \text{Lip}(\Phi; \hat{\mathbf{S}}) & \leq \text{Lip}(\sigma)^{L-1} \text{Lip}(f_{\hat{\mathbf{S}}}^{(1)}) \dots \text{Lip}(f_{\hat{\mathbf{S}}}^{(1)}) \\ & \leq B_\sigma^{L-1} \Pi_L, \end{aligned} \quad (26)$$

where Π_L is a shorthand for $B_1 B_2 \dots B_L$.

For the $\text{Lip}(\Phi; \mathbf{X})$ term, denote the output of the ℓ^{th} layer as $\mathbf{X}_\ell$ when the input is $\mathbf{S}$, and $\hat{\mathbf{X}}_\ell$ otherwise. Additionally, set $\mathbf{X}_0 = \hat{\mathbf{X}}_0 = \mathbf{X}$. On each layer $\ell \in [L-1]$, using (25) on $f^{(\ell)}$, we have

$$\begin{aligned} \|\mathbf{X}_\ell - \hat{\mathbf{X}}_\ell\|_{\text{F}} & = \|\sigma(f^{(\ell)}(\mathbf{X}_{\ell-1}, \mathbf{S})) - \sigma(f^{(\ell)}(\hat{\mathbf{X}}_{\ell-1}, \hat{\mathbf{S}}))\|_{\text{F}} \\ & \leq B_\sigma \|f^{(\ell)}(\mathbf{X}_{\ell-1}, \mathbf{S}) - f^{(\ell)}(\hat{\mathbf{X}}_{\ell-1}, \hat{\mathbf{S}})\|_{\text{F}} \\ & \leq B_\sigma \left(C_\ell \|\mathbf{X}_{\ell-1}\|_{\text{F}} \epsilon_S + B_\ell \|\mathbf{X}_{\ell-1} - \hat{\mathbf{X}}_{\ell-1}\|_{\text{F}} \right). \end{aligned}$$

This also holds for $\ell = L$ without the factor B_σ . We first give a bound for $\|\mathbf{X}_\ell\|_{\text{F}}$ when $\ell \leq L-1$. By properties of σ and triangle inequality,

$$\begin{aligned} \|\mathbf{X}_\ell\|_{\text{F}} & = \|\sigma(f_{\hat{\mathbf{S}}}^{(\ell)}(\mathbf{X}_{\ell-1}))\|_{\text{F}} \\ & = \|\sigma(f_{\hat{\mathbf{S}}}^{(\ell)}(\mathbf{X}_{\ell-1})) - \sigma(0)\|_{\text{F}} \\ & \leq B_\sigma \|f_{\hat{\mathbf{S}}}^{(\ell)}(\mathbf{X}_{\ell-1}) - 0\|_{\text{F}} \\ & \leq B_\sigma \left(\|f_{\hat{\mathbf{S}}}^{(\ell)}(\mathbf{X}_{\ell-1}) - f_{\hat{\mathbf{S}}}^{(\ell)}(0)\|_{\text{F}} + \|f_{\hat{\mathbf{S}}}^{(\ell)}(0)\|_{\text{F}} \right) \\ & \leq B_\sigma (B_\ell \|\mathbf{X}_{\ell-1}\|_{\text{F}} + E_\ell), \end{aligned}$$

where $E_\ell = \|\mathbf{1b}^{(\ell)}\|_{\text{F}} = \sqrt{N} \|\mathbf{b}^{(\ell)}\|_2$. Consider arbitrary real sequences $\{a_n\}, \{b_n\}, \{c_n\} \subset \mathbb{R}$ such that $a_n \leq b_n + c_n a_{n-1}, \forall n \in \{0, 1, 2, \dots\}$, then

$$a_n \leq a_0 c_1 \dots c_n + \sum_{k=1}^n b_k c_{k+1} \dots c_n. \quad (27)$$

Therefore, let $a_\ell = \|\mathbf{X}_\ell\|_{\text{F}}, b_\ell = B_\sigma E_\ell, c_\ell = B_\sigma B_\ell$, we have for $\ell \in [L-1]$

$$\|\mathbf{X}_\ell\|_{\text{F}} \leq B_\sigma^\ell \Pi_\ell \left(n_0 + \sum_{k=1}^{\ell} \frac{E_k}{B_\sigma^{k-1} \Pi_k} \right) =: N_\ell.$$

This holds with $\ell = 0$ as well. Again, let $a'_\ell = \|\mathbf{X}_\ell - \hat{\mathbf{X}}_\ell\|_{\text{F}}, b'_\ell = B_\sigma^{1(\ell < L)} N_{\ell-1} C_{\ell \in \mathcal{S}}, c'_\ell = B_\sigma^{1(\ell < L)} B_\ell$, with boundary $a'_0 = \|\mathbf{X}_0 - \hat{\mathbf{X}}_0\|_{\text{F}} = 0$, we have

$$\begin{aligned} \|\mathbf{X}_L - \hat{\mathbf{X}}_L\|_{\text{F}} & \leq \sum_{\ell=1}^L b'_\ell c'_{\ell+1} \dots c'_L \\ & = N_{L-1} C_L \epsilon_S \\ & \quad + \sum_{\ell=1}^{L-1} B_\sigma N_{\ell-1} C_{\ell \in \mathcal{S}} \cdot B_\sigma^{L-\ell-1} \frac{\Pi_L}{\Pi_\ell} \\ & = \sum_{\ell=1}^L B_\sigma^{L-\ell} \frac{\Pi_L}{\Pi_\ell} C_{\ell \in \mathcal{S}} \cdot N_{\ell-1} \\ & \leq \Pi_L B_\sigma^{L-1} \sum_{\ell=1}^L \frac{C_{\ell \in \mathcal{S}}}{B_\ell} \left(n_0 + \sum_{k=1}^{\ell-1} \frac{E_k}{B_\sigma^{k-1} \Pi_k} \right). \end{aligned}$$

The desired bound is obtained by plugging this and (26) into (25).

B. Proof of Proposition 2

For (a), we bound the output difference in (4). The definition (9) can be written as $\varphi(\mathbf{X}, \mathbf{S}) = (\kappa \circ \pi)(\Phi(\mathbf{X}, \mathbf{S}))$. For fixed $\mathbf{X}$, there is

$$\begin{aligned} \|\varphi(\mathbf{X}, \mathbf{S}) - \varphi(\mathbf{X}, \hat{\mathbf{S}})\|_{\text{F}} & \leq \text{Lip}(\kappa \circ \pi) \|\Phi(\mathbf{X}, \mathbf{S}) - \Phi(\mathbf{X}, \hat{\mathbf{S}})\|_{\text{F}} \\ & \leq \text{Lip}(\kappa) \text{Lip}(\pi) \text{Lip}(\Phi; \mathbf{X}) \|\mathbf{S} - \hat{\mathbf{S}}\|_2 \end{aligned}$$

and the proof goes the same for fixed $\mathbf{S}$.

For (b), using the definition of π ,

$$\begin{aligned} \|\pi(\mathbf{X}) - \pi(\hat{\mathbf{X}})\|_2^2 & = \sum_{p=1}^{F_L} \left| [\pi(\mathbf{X})]_p - [\pi(\hat{\mathbf{X}})]_p \right|^2 \\ & = \sum_{p=1}^{F_L} \left| r(\mathbf{X}_{:,p}) - r(\hat{\mathbf{X}}_{:,p}) \right|^2 \\ & \leq \sum_{p=1}^{F_L} \text{Lip}(r)^2 \|\mathbf{X}_{:,p} - \hat{\mathbf{X}}_{:,p}\|_2^2 \\ & = \text{Lip}(r)^2 \|\mathbf{X} - \hat{\mathbf{X}}\|_{\text{F}}^2. \end{aligned}$$

For the functions *sum* and *mean*, there is

$$|\mathbf{x}^\top \mathbf{1} - \hat{\mathbf{x}}^\top \mathbf{1}| \leq \sum_{i=1}^N |[\mathbf{x}]_i - [\hat{\mathbf{x}}]_i| \leq \sqrt{N} \|\mathbf{x} - \hat{\mathbf{x}}\|_2.$$

For $\max$, let a, b be the maxima in $\mathbf{x}, \hat{\mathbf{x}}$ and i, j be their indices. If $i = j$,

$$|a - b| = |[\mathbf{x}]_i - [\hat{\mathbf{x}}]_i| \leq \|\mathbf{x} - \hat{\mathbf{x}}\|_2.$$

If $i \neq j$, suppose $b \geq a$. Because a is a maximum, $a = [\mathbf{x}]_i \geq [\mathbf{x}]_j$, we have

$$|a - b| = b - a \leq b - [\mathbf{x}]_j \leq \|\mathbf{x} - \hat{\mathbf{x}}\|_2.$$

The case of $b < a$ follows by a symmetric argument.

For (c), r is permutation-invariant if for any permutation matrix $\mathbf{P} \in \mathbb{R}^{n \times n}$ and $\mathbf{x} \in \mathbb{R}^n$, $r(\mathbf{P}\mathbf{x}) = r(\mathbf{x})$. Therefore $[\pi(\mathbf{P}\mathbf{X})]_p = r([\mathbf{P}\mathbf{X}]_{:p}) = r(\mathbf{P}[\mathbf{X}]_{:p}) = r(\mathbf{X}_{:p}) = [\pi(\mathbf{X})]_p$, $\pi(\mathbf{P}\mathbf{X}) = \pi(\mathbf{X})$. A graph isomorphism is given by $(\mathbf{X}, \mathbf{S}) \mapsto (\mathbf{P}\mathbf{X}, \mathbf{P}\mathbf{S}\mathbf{P}^\top)$ for any permutation matrix $\mathbf{P}$. By [25, Proposition 2, without restricting $F_0 = F_L = 1$], for any $\mathbf{X} \in \mathbb{R}^{N \times F_0}$, $\mathbf{S} \in \mathbb{R}^{N \times N}$,

$$\Phi(\mathbf{P}\mathbf{X}, \mathbf{P}\mathbf{S}\mathbf{P}^\top) = \mathbf{P}\Phi(\mathbf{X}, \mathbf{S}).$$

Combining with the permutation invariance of π ,

$$\varphi(\mathbf{P}\mathbf{X}, \mathbf{P}\mathbf{S}\mathbf{P}^\top) = \kappa(\pi(\mathbf{P}\Phi(\mathbf{X}, \mathbf{S}))) = \varphi(\mathbf{X}, \mathbf{S}).$$

C. Proof of Proposition 5

For both perturbations, denote the original output as $\mathbf{Y}$ and the perturbed output as $\hat{\mathbf{Y}}$, and intermediate results as $f_{pq}, \hat{f}_{pq}$.

$$\|\mathbf{Y} - \hat{\mathbf{Y}}\|_F^2 = \sum_{q=1}^Q \left\| \sum_{p=1}^P (f_{pq} - \hat{f}_{pq}) \right\|_2^2.$$

By the definitions (4) and (5), there is

$$\begin{aligned} \|f_{pq} - \hat{f}_{pq}\|_2 &\leq \text{Lip}(f_{pq}; \mathbf{S}) \|\mathbf{X}_{:p} - \hat{\mathbf{X}}_{:p}\|_2, \text{ or} \\ \|f_{pq} - \hat{f}_{pq}\|_2 &\leq \|\mathbf{S} - \hat{\mathbf{S}}\|_2 \overline{\text{Lip}}(f_{pq}) \|\mathbf{X}_{:p}\|_2. \end{aligned}$$

The RHS of both bounds have the form $B_{pq} \|\mathbf{z}_p\|$, with $\mathbf{z}_p = \mathbf{X}_{:p} - \hat{\mathbf{X}}_{:p}$ or $\mathbf{z}_p = \mathbf{X}_{:p}$. Let $B = \max_{p,q} B_{pq}$ and $\mathbf{Z} = [\mathbf{z}_1 | \dots | \mathbf{z}_P]$, we have

$$\begin{aligned} \|\mathbf{Y} - \hat{\mathbf{Y}}\|_F^2 &\leq \sum_{q=1}^Q \left(\sum_{p=1}^P \|f_{pq} - \hat{f}_{pq}\|_2 \right)^2 \\ &\leq \sum_{q=1}^Q \left(\sum_{p=1}^P B_{pq} \|\mathbf{z}_p\|_2 \right)^2 \\ &\leq B^2 \sum_{q=1}^Q \left(\sum_{p=1}^P \|\mathbf{z}_p\|_2 \right)^2 \\ &\leq B^2 \sum_{q=1}^Q \left(\sqrt{P} \|\mathbf{Z}\|_F \right)^2 \\ &= B^2 PQ \|\mathbf{Z}\|_F^2. \end{aligned}$$

Taking square root on both sides, we have $\|\mathbf{Y} - \hat{\mathbf{Y}}\|_F \leq \sqrt{PQB} \|\mathbf{Z}\|_F$, and (14) follows by substituting in the corresponding B and $\mathbf{Z}$.

D. Proof of Proposition 3

Let the adjacency matrix for the original graph be $\mathbf{A} \in \{0, 1\}^{N \times N}$, then the graph with M additional nodes has an adjacency $\mathbf{A} \oplus 0_{M \times M}$. The augmented adjacency becomes $(\mathbf{A} \oplus 0_{M \times M}) + \mathbf{I} = \mathbf{A} \oplus \mathbf{I}_{M \times M}$. Therefore, the degree matrices become $\mathbf{D}' = \mathbf{D} \oplus 0$ and $\tilde{\mathbf{D}}' = \tilde{\mathbf{D}} \oplus \mathbf{I}$, and the claimed form $\mathbf{S} \oplus \mathbf{C}$ for the new GSO can be calculated for each type of GSO using these expressions.

For either $\mathbf{C} = \mathbf{I}$ or $\mathbf{C} = 0$, there is $\mathbf{C}^k = \mathbf{C}$ for any positive integer k . Let f be a GCNN layer function (we omit the index ℓ for brevity). We divide an $(N + M)$ -node input matrix into two matrices with N and M rows, respectively, written as $\mathbf{X}_1 \parallel \mathbf{X}_2$, where $\parallel$ denotes vertical concatenation. Then,

$$\begin{aligned} &f(\mathbf{X}_1 \parallel \mathbf{X}_2, \mathbf{S} \oplus \mathbf{C}) \\ &= \sum_{k \in \mathcal{K}} (\mathbf{S} \oplus \mathbf{C})^k (\mathbf{X}_1 \parallel \mathbf{X}_2) \boldsymbol{\Theta}_k + \mathbf{1}_{N+M} \mathbf{b}^\top \\ &= \sum_{k \in \mathcal{K}} (\mathbf{S}^k \oplus \mathbf{C}) (\mathbf{X}_1 \parallel \mathbf{X}_2) \boldsymbol{\Theta}_k + \mathbf{1}_{N+M} \mathbf{b}^\top \\ &= \sum_{k \in \mathcal{K}} (\mathbf{S}^k \mathbf{X}_1 \parallel \mathbf{C} \mathbf{X}_2) \boldsymbol{\Theta}_k + \mathbf{1}_{N+M} \mathbf{b}^\top \\ &= \left[\sum_{k \in \mathcal{K}} \mathbf{S}^k \mathbf{X}_1 \boldsymbol{\Theta}_k + \mathbf{1}_N \mathbf{b}^\top \parallel \mathbf{C} \mathbf{X}_2 \mathbf{Z} + \mathbf{1}_M \mathbf{b}^\top \right]. \end{aligned}$$

Therefore, a layer function processes the upper and lower parts of its input independently. For the first layer, $\mathbf{X}_1 = \mathbf{X}$ and $\mathbf{X}_2 = \mathbf{X}_+$. The conclusion follows from the definition of GCNN (3).

S2 PROOFS OF SPECTRAL LIPSCHITZ UPPER BOUNDS

We reformulate (18) into a tensor contraction

$$\tilde{\mathbf{Y}} = \langle \tilde{\mathbf{X}}, \mathcal{H} \rangle \quad (28)$$

where $[\mathcal{H}]_{ipjq} = [\mathbf{H}(\lambda_i)]_{pq} \mathbb{1}_{i=j}$, so the bilinearity property of the tensor contraction can be applied.

A. Proof of Proposition 8

We substitute (28) into the definition of signal Lipschitz constant (4) and transform it with orthogonal matrices and tensor re-indexing:

$$\begin{aligned} \text{Lip}(\tau; \mathbf{S}) &= \sup_{\mathbf{x} \neq \hat{\mathbf{x}}} \frac{\|\tau(\mathbf{X}, \mathbf{S}) - \tau(\hat{\mathbf{X}}, \mathbf{S})\|_F}{\|\mathbf{X} - \hat{\mathbf{X}}\|_F} \\ &= \sup_{\mathbf{x} \neq \hat{\mathbf{x}}} \frac{\|\mathbf{V} \langle \mathbf{V}^\top (\mathbf{X} - \hat{\mathbf{X}}), \mathcal{H} \rangle\|_F}{\|\mathbf{V}^\top (\mathbf{X} - \hat{\mathbf{X}})\|_F} \\ &= \sup_{\mathbf{D} \in \mathbb{R}^{N \times P}} \frac{\|\langle \mathbf{D}, \mathcal{H} \rangle\|_F}{\|\mathbf{D}\|_F} \\ &= \sup_{\mathbf{v} \in \mathbb{R}^{NP}} \frac{\|\mathcal{H}_{(2)}^\top \mathbf{v}\|_2}{\|\mathbf{v}\|_2} \\ &= \|\mathcal{H}_{(2)}\|_2 \\ &= \|\mathbf{H}(\lambda_1) \oplus \dots \oplus \mathbf{H}(\lambda_N)\|_2 \\ &= \max_{i \in [N]} \|\mathbf{H}(\lambda_i)\|_2. \end{aligned} \quad (29)$$

Here, $\mathcal{H}_{(2)}$ is the 2-unfolding [68] of the tensor $\mathcal{H}$, which satisfies $\text{vec}(\langle \mathbf{X}, \mathcal{H} \rangle) = \mathcal{H}_{(2)}^T \text{vec}(\mathbf{X})$, and $\oplus$ means concatenating diagonally.

B. Proof of Proposition 9

By Lemma 7, for any eigenbases $\mathbf{V}$ of $\mathbf{S}$ and $\hat{\mathbf{V}}$ of $\hat{\mathbf{S}}$, there is

$$\begin{aligned} & \left\| \tau(\mathbf{X}, \mathbf{S}) - \tau(\mathbf{X}, \hat{\mathbf{S}}) \right\|_{\text{F}} \\ &= \left\| \mathbf{V} \langle \mathbf{V}^T \mathbf{X}, \mathcal{H} \rangle - \hat{\mathbf{V}} \langle \hat{\mathbf{V}}^T \mathbf{X}, \hat{\mathcal{H}} \rangle \right\|_{\text{F}} \\ &\leq \left\| \hat{\mathbf{V}} \langle \hat{\mathbf{V}}^T \mathbf{X}, \mathcal{H} - \hat{\mathcal{H}} \rangle \right\|_{\text{F}} \end{aligned} \quad (30)$$

$$+ \left\| \mathbf{V} \langle \mathbf{V}^T \mathbf{X}, \mathcal{H} \rangle - \hat{\mathbf{V}} \langle \hat{\mathbf{V}}^T \mathbf{X}, \mathcal{H} \rangle \right\|_{\text{F}}. \quad (31)$$

For the first term (30), we define an auxiliary function g on the basis $\hat{\mathbf{V}}$ with frequency response $\mathbf{H}^-(\hat{\lambda}_i) = \mathbf{H}(\lambda_i) - \mathbf{H}(\hat{\lambda}_i)$. By (28), g can be written as $g(\mathbf{X}, \mathbf{S}) = \hat{\mathbf{V}} \langle \hat{\mathbf{V}}^T \mathbf{X}, \mathcal{H} - \hat{\mathcal{H}} \rangle$. Therefore, by Proposition 8,

$$\begin{aligned} & \left\| \hat{\mathbf{V}} \langle \hat{\mathbf{V}}^T \mathbf{X}, \mathcal{H} - \hat{\mathcal{H}} \rangle \right\|_{\text{F}} \\ &= \|g(\mathbf{X}, \mathbf{S}) - g(0, \mathbf{S})\|_{\text{F}} \\ &\leq \|\mathbf{X}\|_{\text{F}} \text{Lip}(g; \mathbf{S}) \\ &= \|\mathbf{X}\|_{\text{F}} \max_{i \in [N]} \left\| \mathbf{H}(\lambda_i) - \mathbf{H}(\hat{\lambda}_i) \right\|_2 \\ &= \|\mathbf{X}\|_{\text{F}} \max_{i \in [N]} \left\| \int_{\hat{\lambda}_i}^{\lambda_i} \mathbf{H}'(\lambda) d\lambda \right\|_2 \\ &\leq \|\mathbf{X}\|_{\text{F}} \max_{i \in [N]} |\lambda_i - \hat{\lambda}_i| \sup_{\lambda \in [\lambda_i, \hat{\lambda}_i]} \|\mathbf{H}'(\lambda)\|_2 \\ &\leq \|\mathbf{X}\|_{\text{F}} \|\mathbf{E}\|_2 \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}'(\lambda)\|_2, \end{aligned}$$

where the last step follows from Weyl's inequality [67, Cor. 8.1.6]. The second term (31) can be decomposed as

$$\begin{aligned} & \left\| \mathbf{V} \langle \mathbf{V}^T \mathbf{X}, \mathcal{H} \rangle - \hat{\mathbf{V}} \langle \hat{\mathbf{V}}^T \mathbf{X}, \mathcal{H} \rangle \right\|_{\text{F}} \\ &\leq \left\| \mathbf{V} \langle (\mathbf{V} - \hat{\mathbf{V}})^T \mathbf{X}, \mathcal{H} \rangle \right\|_{\text{F}} + \left\| (\mathbf{V} - \hat{\mathbf{V}}) \langle \hat{\mathbf{V}}^T \mathbf{X}, \mathcal{H} \rangle \right\|_{\text{F}} \\ &\leq 2 \|\mathbf{X}\|_{\text{F}} \|\mathbf{V} - \hat{\mathbf{V}}\|_2 \text{Lip}(\langle \cdot, \mathcal{H} \rangle). \end{aligned}$$

By (29), there is $\text{Lip}(\langle \cdot, \mathcal{H} \rangle) = \max_{\lambda \in \lambda(\mathbf{S})} \|\mathbf{H}(\lambda)\|_2 \leq \max_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}(\lambda)\|_2$. Finally, if $\mathbf{S}$ and $\hat{\mathbf{S}}$ do not share the same eigenbasis, there is $\|\mathbf{V} - \hat{\mathbf{V}}\|_2 \leq \|\mathbf{V}\|_2 + \|\hat{\mathbf{V}}\|_2 = 2$.

C. Proof of Proposition 10

Let the original and perturbed GSOs be $\mathbf{S} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$ and $\hat{\mathbf{S}} = \hat{\mathbf{V}} \hat{\mathbf{\Lambda}} \hat{\mathbf{V}}^T$, and let each frequency component of the signal input under the two eigenbases be $\mathbf{x}_i = [\mathbf{V}^T \mathbf{X}]_{i:}$ and $\hat{\mathbf{x}}_i = [\hat{\mathbf{V}}^T \mathbf{X}]_{i:}$ (as row vectors). We consider the output difference $\tau(\mathbf{X}, \mathbf{S}) - \tau(\mathbf{X}, \hat{\mathbf{S}})$ in the Fourier domain corresponding to one eigenbasis, say $\mathbf{V}$, also by frequency component:

$$\mathbf{d}_i := \left[\mathbf{V}^T (\tau(\mathbf{X}, \mathbf{S}) - \tau(\mathbf{X}, \hat{\mathbf{S}})) \right]_{i:},$$

so that the squared Frobenius norm of the output difference can be written as

$$\begin{aligned} \left\| \tau(\mathbf{X}, \mathbf{S}) - \tau(\mathbf{X}, \hat{\mathbf{S}}) \right\|_{\text{F}}^2 &= \left\| \mathbf{V}^T (\tau(\mathbf{X}, \mathbf{S}) - \tau(\mathbf{X}, \hat{\mathbf{S}})) \right\|_{\text{F}}^2 \\ &= \sum_{i=1}^N \|\mathbf{d}_i\|_2^2. \end{aligned}$$

By (18), the perturbed term in $\mathbf{d}_i$ can be rewritten as

$$\begin{aligned} [\mathbf{V}^T \tau(\mathbf{X}, \hat{\mathbf{S}})]_{i:} &= [\mathbf{V}^T \hat{\mathbf{V}} \hat{\mathbf{V}}^T \tau(\mathbf{X}, \hat{\mathbf{S}})]_{i:} \\ &= [\mathbf{V}^T \hat{\mathbf{V}}]_{i:} \hat{\mathbf{V}}^T \tau(\mathbf{X}, \hat{\mathbf{S}}) \\ &= \sum_{j=1}^N [\mathbf{V}^T \hat{\mathbf{V}}]_{ij} [\hat{\mathbf{V}}^T \tau(\mathbf{X}, \hat{\mathbf{S}})]_{j:} \\ &= \sum_{j=1}^N [\mathbf{V}^T \hat{\mathbf{V}}]_{ij} \hat{\mathbf{x}}_j \mathbf{H}(\hat{\lambda}_j). \end{aligned}$$

In the following, we denote the matrix $\mathbf{V}^T \hat{\mathbf{V}}$ as $\mathbf{W}$. Next, rewrite the original term $[\mathbf{V}^T \tau(\mathbf{X}, \mathbf{S})]_{i:} = \mathbf{x}_i \mathbf{H}(\lambda_i)$ also as a sum over $\hat{\mathbf{x}}_j$. Observe that $\mathbf{V}^T \mathbf{X} = \mathbf{V}^T \hat{\mathbf{V}} \hat{\mathbf{V}}^T \mathbf{X} = \mathbf{W} \hat{\mathbf{V}}^T \mathbf{X}$, so $\mathbf{x}_i = [\mathbf{W}]_{i:} \hat{\mathbf{V}}^T \mathbf{X} = \sum_{j=1}^N [\mathbf{W}]_{ij} \hat{\mathbf{x}}_j$. Therefore,

$$\begin{aligned} \mathbf{d}_i &= [\mathbf{V}^T \tau(\mathbf{X}, \mathbf{S})]_{i:} - [\mathbf{V}^T \tau(\mathbf{X}, \hat{\mathbf{S}})]_{i:} \\ &= \sum_{j=1}^N [\mathbf{W}]_{ij} \hat{\mathbf{x}}_j \mathbf{H}(\hat{\lambda}_j) - \sum_{j=1}^N [\mathbf{W}]_{ij} \hat{\mathbf{x}}_j \mathbf{H}(\lambda_i) \\ &= \sum_{j=1}^N [\mathbf{W}]_{ij} \hat{\mathbf{x}}_j (\mathbf{H}(\hat{\lambda}_j) - \mathbf{H}(\lambda_i)). \end{aligned}$$

By the triangle inequality and Cauchy-Schwarz inequality, as well as the bounding technique for differences in $\mathbf{H}(\lambda)$ used in Section S2-B,

$$\begin{aligned} \|\mathbf{d}_i\|_2 &\leq \sum_{j=1}^N |[\mathbf{W}]_{ij}| \left\| \hat{\mathbf{x}}_j (\mathbf{H}(\hat{\lambda}_j) - \mathbf{H}(\lambda_i)) \right\|_2 \\ &\leq \sum_{j=1}^N |[\mathbf{W}]_{ij}| \|\hat{\mathbf{x}}_j\|_2 \left\| \mathbf{H}(\hat{\lambda}_j) - \mathbf{H}(\lambda_i) \right\|_2 \\ &\leq \sum_{j=1}^N |[\mathbf{W}]_{ij}| \|\hat{\mathbf{x}}_j\|_2 |\hat{\lambda}_j - \lambda_i| C_H \\ &\leq C_H \left(\sum_{j=1}^N \|\hat{\mathbf{x}}_j\|_2^2 \right)^{1/2} \left(\sum_{j=1}^N [\mathbf{W}]_{ij}^2 (\hat{\lambda}_j - \lambda_i)^2 \right)^{1/2} \\ &= C_H \|\mathbf{X}\|_{\text{F}} \left(\sum_{j=1}^N [\mathbf{W}]_{ij}^2 (\hat{\lambda}_j - \lambda_i)^2 \right)^{1/2}. \end{aligned}$$

where $C_H = \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}'(\lambda)\|_2$. Observe that

$$\begin{aligned} [\mathbf{W}]_{ij} (\hat{\lambda}_j - \lambda_i) &= [\mathbf{W} \hat{\mathbf{\Lambda}} - \mathbf{\Lambda} \mathbf{W}]_{ij} \\ &= [\mathbf{V}^T \hat{\mathbf{V}} \hat{\mathbf{\Lambda}} - \mathbf{\Lambda} \mathbf{V}^T \hat{\mathbf{V}}]_{ij} \\ &= [\mathbf{V}^T \hat{\mathbf{V}} \hat{\mathbf{\Lambda}} \hat{\mathbf{V}}^T \hat{\mathbf{V}} - \mathbf{V}^T \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T \hat{\mathbf{V}}]_{ij} \\ &= [\mathbf{V}^T (\hat{\mathbf{S}} - \mathbf{S}) \hat{\mathbf{V}}]_{ij}. \end{aligned}$$

Finally,

$$\begin{aligned} \sum_{i=1}^N \|\mathbf{d}_i\|_2^2 &\leq \sum_{i=1}^N C_H^2 \|\mathbf{X}\|_F^2 \sum_{j=1}^N [\mathbf{W}]_{ij}^2 (\hat{\lambda}_j - \lambda_i)^2 \\ &= C_H^2 \|\mathbf{X}\|_F^2 \sum_{i=1}^N \sum_{j=1}^N [\mathbf{V}^\top (\hat{\mathbf{S}} - \mathbf{S}) \hat{\mathbf{V}}]_{ij}^2 \\ &= C_H^2 \|\mathbf{X}\|_F^2 \|\mathbf{S} - \hat{\mathbf{S}}\|_F^2, \end{aligned}$$

hence

$$\|\tau(\mathbf{X}, \mathbf{S}) - \tau(\mathbf{X}, \hat{\mathbf{S}})\|_F \leq C_H \|\mathbf{X}\|_F \|\mathbf{S} - \hat{\mathbf{S}}\|_F.$$

To show the equality $\overline{\text{Lip}}_F(\tau) = C_H$, we need to find $\mathbf{X}, \mathbf{S}, \hat{\mathbf{S}}$ so that the above inequality is attained. Because $\|\mathbf{H}'(\lambda)\|$ is continuous, the supremum in C_H is attained by some λ_0 . Let $\mathbf{U} \in \mathbb{R}^{N \times N}$ be an orthogonal matrix and $\mathbf{u} = \mathbf{U}_{:,1}$. Let $\mathbf{v} \in \mathbb{R}^P$ be a unit vector maximizing $\|\mathbf{v}^\top \mathbf{H}'(\lambda_0)\|_2$. Take $\mathbf{X} = \mathbf{u}\mathbf{v}^\top$, $\mathbf{S} = \lambda_0 \mathbf{I}$, and $\hat{\mathbf{S}} = \mathbf{S} + \epsilon \mathbf{u}\mathbf{u}^\top = \mathbf{U} \text{diag}(\lambda_0 + \epsilon, \lambda_0, \dots, \lambda_0) \mathbf{U}^\top$. Because $\lambda(\hat{\mathbf{S}}) = \{\lambda_0, \lambda_0 + \epsilon\}$, there must be some $\mu > 0$ such that with $I_\mu = (0, \mu)$ or $I_\mu = (-\mu, 0)$, for any $\epsilon \in I_\mu$, there is $\lambda(\hat{\mathbf{S}}) \subset [\underline{\lambda}, \bar{\lambda}]$, so that $\hat{\mathbf{S}} \in \mathcal{S}^*$. We have $\|\mathbf{X}\|_F^2 = \sum_{i=1}^N \sum_{p=1}^P ([\mathbf{u}]_i [\mathbf{v}]_p)^2 = \sum_{i=1}^N [\mathbf{u}]_i^2 \sum_{p=1}^P [\mathbf{v}]_p^2 = 1$, and $\|\mathbf{S} - \hat{\mathbf{S}}\|_F = \epsilon \|\mathbf{u}\mathbf{u}^\top\|_F = \epsilon$, similarly. Under the common eigenbasis $\mathbf{U}$ of $\mathbf{S}$ and $\hat{\mathbf{S}}$, $\tilde{\mathbf{X}} = \mathbf{U}^\top \mathbf{X} = \mathbf{U}^\top \mathbf{u}\mathbf{v}^\top = \mathbf{e}_1 \mathbf{v}^\top$, so $\tilde{\mathbf{Y}}_i = \tilde{\mathbf{X}}_i: \mathbf{H}(\lambda_i) = \mathbb{1}_{i=1} \mathbf{v}^\top \mathbf{H}(\lambda_1)$ for both GSOs. We have $\|\mathbf{Y} - \hat{\mathbf{Y}}\|_F = \|\tilde{\mathbf{Y}} - \hat{\tilde{\mathbf{Y}}}\|_F = \|\mathbf{v}^\top (\mathbf{H}(\lambda_0) - \mathbf{H}(\lambda_0 + \epsilon))\|_2$. Therefore,

$$\begin{aligned} &\sup_{\mathbf{X}, \mathbf{S}, \hat{\mathbf{S}}} \frac{\|\tau(\mathbf{X}, \mathbf{S}) - \tau(\mathbf{X}, \hat{\mathbf{S}})\|_F}{\|\mathbf{X}\|_F \|\mathbf{S} - \hat{\mathbf{S}}\|_F} \\ &\geq \sup_{\epsilon \in I_\mu} \frac{\|\mathbf{v}^\top (\mathbf{H}(\lambda_0) - \mathbf{H}(\lambda_0 + \epsilon))\|_2}{\epsilon} \\ &\geq \lim_{\epsilon \rightarrow 0} \left\| \mathbf{v}^\top \frac{\mathbf{H}(\lambda_0) - \mathbf{H}(\lambda_0 + \epsilon)}{\epsilon} \right\|_2 \\ &= \|\mathbf{v}^\top \mathbf{H}'(\lambda_0)\|_2 \\ &= \|\mathbf{H}'(\lambda_0)\|_2 \\ &= C_H. \end{aligned}$$

Finally, because for any $\mathbf{S} \neq \hat{\mathbf{S}}$, there is $1/\|\mathbf{S} - \hat{\mathbf{S}}\|_2 \leq \sqrt{N}/\|\mathbf{S} - \hat{\mathbf{S}}\|_F$, $\overline{\text{Lip}}(\tau) \leq \sqrt{N} \overline{\text{Lip}}_F(\tau)$ by the monotonicity of supremum.

S3 PROOFS OF CONVOLUTIONAL LIPSCHITZ BOUNDS

A. Proof of Proposition 6

The signal Lipschitz constant can be bounded by the fact that matrix norms are subadditive and submultiplicative, and by the inequality $\|\mathbf{AB}\|_F \leq \|\mathbf{A}\|_2 \|\mathbf{B}\|_F$.

$$\begin{aligned} \|f(\mathbf{X}, \mathbf{S}) - f(\hat{\mathbf{X}}, \mathbf{S})\|_F &= \left\| \sum_{k \in \mathcal{K}} \mathbf{S}^k (\mathbf{X} - \hat{\mathbf{X}}) \boldsymbol{\Theta}_k \right\|_F \\ &\leq \sum_{k \in \mathcal{K}} \|\mathbf{S}\|_2^k \|\mathbf{X} - \hat{\mathbf{X}}\|_F \|\boldsymbol{\Theta}_k\|_2 \\ &\leq \|\mathbf{X} - \hat{\mathbf{X}}\|_F \sum_{k \in \mathcal{K}} \bar{s}^k \|\boldsymbol{\Theta}_k\|_2. \end{aligned}$$

For the structure Lipschitz constant, observe that $\mathbf{S}^k - \hat{\mathbf{S}}^k = \sum_{j=0}^{k-1} \mathbf{S}^j (\mathbf{S} - \hat{\mathbf{S}}) \hat{\mathbf{S}}^{k-1-j}$, so we have

$$\begin{aligned} &\|f(\mathbf{X}, \mathbf{S}) - f(\mathbf{X}, \hat{\mathbf{S}})\|_F \\ &= \left\| \sum_{k \in \mathcal{K}} (\mathbf{S}^k - \hat{\mathbf{S}}^k) \mathbf{X} \boldsymbol{\Theta}_k \right\|_F \\ &\leq \sum_{k \in \mathcal{K}} \|\mathbf{S}^k - \hat{\mathbf{S}}^k\|_2 \|\mathbf{X}\|_F \|\boldsymbol{\Theta}_k\|_2 \\ &\leq \|\mathbf{X}\|_F \sum_{k \in \mathcal{K}} \|\boldsymbol{\Theta}_k\|_2 \sum_{j=0}^{k-1} \|\mathbf{S}^j (\mathbf{S} - \hat{\mathbf{S}}) \hat{\mathbf{S}}^{k-1-j}\|_2 \\ &\leq \|\mathbf{X}\|_F \|\mathbf{S} - \hat{\mathbf{S}}\|_2 \sum_{k \in \mathcal{K}} k \bar{s}^{k-1} \|\boldsymbol{\Theta}_k\|_2. \end{aligned}$$

B. Proof of Proposition 12

We first show (24) by considering the GSO inputs $\mathbf{S} = \lambda \mathbf{I}$ and $\hat{\mathbf{S}} = \mu \mathbf{I}$, with $\underline{\lambda} < \lambda < \mu < \bar{\lambda}$. By definition,

$$\begin{aligned} \text{Lip}(\gamma; \mathbf{X}) &\geq \sup_{\mu > \lambda} \frac{\|\gamma(\mathbf{X}, \mu \mathbf{I}) - \gamma(\mathbf{X}, \lambda \mathbf{I})\|_F}{\|\mu \mathbf{I} - \lambda \mathbf{I}\|_2} \\ &= \sup_{\mu > \lambda} \left\| \sum_{k \in \mathcal{K}} \frac{\mu^k - \lambda^k}{\mu - \lambda} \mathbf{X} \boldsymbol{\Theta}_k \right\|_F \end{aligned}$$

Using the identity

$$\mu^k - \lambda^k = (\mu - \lambda)(\mu^{k-1} + \mu^{k-2}\lambda + \dots + \lambda^{k-1}),$$

and the continuity of matrix norms,

$$\begin{aligned} &\lim_{\mu \rightarrow \lambda^+} \left\| \sum_{k \in \mathcal{K}} \frac{\mu^k - \lambda^k}{\mu - \lambda} \mathbf{X} \boldsymbol{\Theta}_k \right\|_F \\ &= \left\| \sum_{k \in \mathcal{K}} \left(\lim_{\mu \rightarrow \lambda^+} \sum_{j=0}^{k-1} \mu^j \lambda^{k-1-j} \right) \mathbf{X} \boldsymbol{\Theta}_k \right\|_F \\ &= \left\| \sum_{k \in \mathcal{K}} k \lambda^{k-1} \mathbf{X} \boldsymbol{\Theta}_k \right\|_F \\ &= \|\mathbf{X} \mathbf{H}'(\lambda)\|_F. \end{aligned}$$

Therefore,

$$\text{Lip}(\gamma; \mathbf{X}) \geq \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{X} \mathbf{H}'(\lambda)\|_F.$$

We further have

$$\begin{aligned} \overline{\text{Lip}}(\gamma) &= \sup_{\mathbf{X} \neq 0} \frac{\text{Lip}(\gamma; \mathbf{X})}{\|\mathbf{X}\|_F} \\ &\geq \sup_{\mathbf{X} \neq 0} \frac{1}{\|\mathbf{X}\|_F} \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{X} \mathbf{H}'(\lambda)\|_F \\ &= \sup_{\mathbf{X} \neq 0} \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \frac{\|\mathbf{X} \mathbf{H}'(\lambda)\|_F}{\|\mathbf{X}\|_F} \\ &= \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \sup_{\mathbf{X} \neq 0} \frac{\|\mathbf{X} \mathbf{H}'(\lambda)\|_F}{\|\mathbf{X}\|_F} \\ &= \sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}'(\lambda)\|_2. \end{aligned}$$

The last step is because the inequality $\|\mathbf{A}\mathbf{B}\|_F \leq \|\mathbf{A}\|_F \|\mathbf{B}\|_2$ achieves equality when $\mathbf{A} = \mathbf{1}\mathbf{v}^\top$, where $\mathbf{v}$ is the left singular vector of $\mathbf{B}$ corresponding to the largest singular value.

To give an example where the strict inequality holds, we consider an example of a single GCF with a frequency response

$$h(\lambda) = \frac{3}{4}\lambda - \frac{3}{8}\lambda^2 - \frac{1}{4}\lambda^3 - \frac{1}{16}\lambda^4.$$

Taking the derivative,

$$h'(\lambda) = 1 - \frac{(\lambda + 1)^3}{4},$$

so when $\bar{\rho} = 1$, $\sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{H}'(\lambda)\|_2 = \sup_{|\lambda| \leq 1} |h'(\lambda)| = 1$.

Therefore, the strict inequality calls for some $\mathbf{x}, \mathbf{S}, \hat{\mathbf{S}}$ such that

$$\|\mathbf{h}(\mathbf{S})\mathbf{x} - \mathbf{h}(\hat{\mathbf{S}})\mathbf{x}\|_2 > \|\mathbf{x}\|_2 \|\mathbf{S} - \hat{\mathbf{S}}\|_2. \quad (32)$$

$$\text{Set } \mathbf{S} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and } \hat{\mathbf{S}} = \mathbf{S} + \delta \mathbf{E}_0, \text{ where } \mathbf{E}_0 = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$$

and $\delta = \sqrt{5}/100$. It is easy to verify that $\mathbf{S}$ and $\hat{\mathbf{S}}$ are symmetric and have spectral radii at most 1. Setting $\mathbf{x} = [1, 1]^\top / \sqrt{2}$, the RHS of (32) becomes $\delta \|\mathbf{x}\|_2 \|\mathbf{E}_0\|_2 = 0.05$, while LHS is greater than 0.054, satisfying the requirement.

S4 PROOF OF LEMMA 7

Let the distinct eigenvalues of $\mathbf{S}$ be $\mu_1 > \dots > \mu_n$, $n \leq N$, where μ_j has multiplicity m_j . Partition $\mathbf{U}, \mathbf{V}$ into eigenvectors corresponding to the same eigenvalue as $\mathbf{U} = [\mathbf{U}_1 | \dots | \mathbf{U}_n]$ and $\mathbf{V} = [\mathbf{V}_1 | \dots | \mathbf{V}_n]$, so that the columns of $\mathbf{U}_j$ or $\mathbf{V}_j$ form orthonormal bases for the eigenspace $E(\mu_j, \mathbf{S}) = \ker(\mu_j \mathbf{I} - \mathbf{S})$. There are orthogonal change-of-basis matrices $\mathbf{W}_j \in \mathbb{R}^{m_j \times m_j}$ such that $\mathbf{V}_j = \mathbf{U}_j \mathbf{W}_j$. Denote $\boldsymbol{\theta} = [h(\lambda_1), \dots, h(\lambda_N)]^\top$ so that we can write the map as a sum

$$\begin{aligned} \mathbf{y} &= \mathbf{V} \text{diag}(\boldsymbol{\theta}) \mathbf{V}^\top \mathbf{x} \\ &= \sum_{i=1}^N h(\lambda_i) \mathbf{v}_i \mathbf{v}_i^\top \mathbf{x} \\ &= \sum_{j=1}^n h(\mu_j) \sum_{k=1}^{m_j} \mathbf{v}_{j,k} \mathbf{v}_{j,k}^\top \mathbf{x} \\ &= \sum_{j=1}^n h(\mu_j) \mathbf{V}_j \mathbf{V}_j^\top \mathbf{x} \\ &= \sum_{j=1}^n h(\mu_j) \mathbf{U}_j \mathbf{U}_j^\top \mathbf{x} \\ &= \mathbf{U} \text{diag}(\boldsymbol{\theta}) \mathbf{U}^\top \mathbf{x}, \end{aligned}$$

where $\mathbf{v}_i$ is the i -th column of $\mathbf{V}$ and $\mathbf{v}_{j,k}$ is the k -th column of $\mathbf{V}_j$.

S5 PROOF OF PROPOSITION 11

For the comparison, we denote $\sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{A}(\lambda)\|_{\max}$ as $\lceil \mathbf{A} \rceil$ and $\sup_{\lambda \in [\underline{\lambda}, \bar{\lambda}]} \|\mathbf{A}(\lambda)\|_2$ as $\langle \mathbf{A} \rangle$. Firstly, we show that assuming $\text{Lip}(\sigma) \leq 1$, Theorems 1 and 4 (and their proofs) in [25] give the following stability bound under the finite spectrum interpretation: for any GCNN node embedding function

Φ without bias terms and inputs $\mathbf{X} \in \mathbb{R}^{N \times F_0}$, $\mathbf{S}, \hat{\mathbf{S}} \in \mathcal{S}^*$, there is $\|\Phi(\mathbf{X}, \mathbf{S}) - \Phi(\mathbf{X}, \mathbf{S} + \mathbf{E})\|_F \leq C_0 \|\mathbf{X}\|_F \|\mathbf{E}\|_2 + O(\|\mathbf{E}\|^2)$ as $\|\mathbf{E}\| \rightarrow 0$, where

$$C_0 = (1 + 8\sqrt{N})LB^{L-1}C \prod_{\ell=1}^L \sqrt{F_\ell F_{\ell-1}}, \quad (33)$$

$B = \max_\ell \lceil \mathbf{H}_\ell \rceil$, $C = \max_\ell \lceil \mathbf{H}'_\ell \rceil$, where $\mathbf{H}_\ell(\lambda) = \sum_{k \in \mathcal{K}_\ell} \lambda^k \boldsymbol{\Theta}_k^{(\ell)}$ is the frequency response of the mGCF on the ℓ -th layer. This is an asymptotic bound that is weaker than $\overline{\text{Lip}}(\Phi) \leq C_0$. The original bound given by [25, Eq.(89) and Theorem 1] can be written as

$$\begin{aligned} &\|\Phi(\mathbf{X}, \hat{\mathbf{S}}) - \Phi(\mathbf{X}, \mathbf{S})\|_F \\ &\leq \sqrt{F_L} \Delta \epsilon (B_\sigma B)^{L-1} \sum_{\ell=1}^L B_\sigma^\ell \cdot \prod_{\ell=1}^{L-1} F_\ell \cdot \sum_{p=1}^{F_0} \|\mathbf{X}_{:,p}\|_2 + O(\epsilon^2) \end{aligned}$$

where $\Delta = C(1 + \delta\sqrt{N})$, $\epsilon = \|\mathbf{S} - \hat{\mathbf{S}}\|_2$, under the following conditions: each GCF in each layer of Φ (cf. (12)) is assumed to satisfy $|h(\lambda_1) - h(\lambda_2)| \leq C|\lambda_1 - \lambda_2|$, and let the filter operators be $\hat{\mathbf{H}}_\ell^{pq} = \sum_{k \in \mathcal{K}} [\boldsymbol{\Theta}_k^{(\ell)}]_{pq} \hat{\mathbf{S}}^k$, there is $\|\hat{\mathbf{H}}_\ell^{pq}\|_2 \leq B$ for all ℓ, p, q . We rewrite the bound with the frequency domain $[\underline{\lambda}, \bar{\lambda}]$ and mGCF notation. The first condition can be written as $|h'(\lambda)| \leq C$. With explicit indexing, it is $|\lceil \mathbf{H}'_\ell(\lambda) \rceil_{pq}| \leq C$, or $\|\mathbf{H}'_\ell(\lambda)\|_{\max} \leq C$, for $\underline{\lambda} \leq \lambda \leq \bar{\lambda}$, that is, $\lceil \mathbf{H}'_\ell \rceil \leq C$. For the B -condition, for any ℓ, p, q , write $\theta_k = [\boldsymbol{\Theta}_k^{(\ell)}]_{pq}$, so

$$\begin{aligned} \|\hat{\mathbf{H}}_\ell^{pq}\|_2 &= \left\| \sum_{k \in \mathcal{K}} \theta_k \hat{\mathbf{V}} \hat{\mathbf{\Lambda}}^k \hat{\mathbf{V}}^\top \right\|_2 \\ &= \left\| \sum_{k \in \mathcal{K}} \theta_k \hat{\mathbf{\Lambda}}^k \right\|_2 \\ &= \left\| \text{diag}_i(\theta_0 + \theta_1 \hat{\lambda}_i + \theta_2 \hat{\lambda}_i^2 + \dots) \right\|_2 \\ &= \max_{i \in [N]} |h_\theta(\hat{\lambda}_i)| \\ &= \max_{i \in [N]} |\lceil \mathbf{H}_\ell(\hat{\lambda}_i) \rceil_{pq}|. \end{aligned}$$

Therefore, $\|\hat{\mathbf{H}}_\ell^{pq}\|_2 \leq B$ is equivalent to

$$\max_{i,p,q,\ell} |\lceil \mathbf{H}_\ell(\hat{\lambda}_i) \rceil_{pq}| = \max_{\ell,i} \|\mathbf{H}_\ell(\hat{\lambda}_i)\|_{\max} \leq B,$$

which should be relaxed as $\lceil \mathbf{H}_\ell \rceil \leq B$ since the spectrum of $\hat{\mathbf{S}}$ is unknown. The remaining terms satisfy $\delta \leq 8$ and $\sum_{p=1}^{F_0} \|\mathbf{X}_{:,p}\|_2 \leq \sqrt{F_0} \|\mathbf{X}\|_F$.

Meanwhile, substituting (22) and (23) into (7), we have

$$\overline{\text{Lip}}(\Phi) \leq \sqrt{N} \sum_{\ell=1}^L \langle \mathbf{H}'_\ell \rangle \prod_{j \neq \ell} \langle \mathbf{H}_j \rangle =: C_1.$$

For any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, there is

$$\|\mathbf{A}\|_{\max} \leq \|\mathbf{A}\|_2 \leq \sqrt{mn} \|\mathbf{A}\|_{\max},$$

so $\lceil \mathbf{H}_\ell \rceil \leq \langle \mathbf{H}_\ell \rangle \leq \sqrt{F_\ell F_{\ell-1}} \lceil \mathbf{H}_\ell \rceil$ and similarly for $\mathbf{H}'_\ell$. Consequently, $\langle \mathbf{H}_\ell \rangle \leq \sqrt{F_\ell F_{\ell-1}} \lceil \mathbf{H}_\ell \rceil \leq \sqrt{F_\ell F_{\ell-1}} B$ and $\langle \mathbf{H}'_\ell \rangle \leq \sqrt{F_\ell F_{\ell-1}} C$. Therefore,

$$\begin{aligned} C_1 &\leq \sqrt{N} \sum_{\ell=1}^L \sqrt{F_\ell F_{\ell-1}} C \prod_{j \neq \ell} \sqrt{F_j F_{j-1}} B \\ &= LCB^{L-1} \sqrt{N} \prod_{\ell=1}^L \sqrt{F_\ell F_{\ell-1}} \\ &\leq C_0. \end{aligned}$$

Moreover, if for all layers $\mathcal{K} = \{1\}$ and $\Theta_1 = \mathbf{I}$, $\mathbf{H}_\ell(\lambda) = \lambda \mathbf{I}$ and $\mathbf{H}'_\ell(\lambda) = \mathbf{I}$ for each ℓ . For any $\bar{\lambda} = -\underline{\lambda} > 0$, there is $\langle \mathbf{H}_\ell \rangle = \lceil \mathbf{H}_\ell \rceil = \bar{\lambda}$, $\langle \mathbf{H}'_\ell \rangle = \lceil \mathbf{H}'_\ell \rceil = 1$, so $C_1 = L\bar{\lambda}^{L-1} \sqrt{N}$, while $C_0 = L\bar{\lambda}^{L-1} (1 + 8\sqrt{N}) \prod_{\ell} \sqrt{F_\ell F_{\ell-1}}$.

S6 PSEUDOCODE FOR THE PGD ATTACK

The PGD attack on $\mathbf{S}$ for maximizing the output deviation for a given spectral norm budget within the domain $\mathcal{S}^*$ is given in Algorithm 1. In our implementation, the spectral norm is computed by a fixed number of power iterations.

Algorithm 1: A PGD attack for empirical Lipschitz lower bound

Data: number of steps $T \in \{0, 1, 2, \dots\}$, step size $\alpha > 0$, target norm n_t , initial GSO $\mathbf{S} \in \mathbb{R}^{N \times N}$, input signals $\mathbf{X} \in \mathbb{R}^{N \times F_0}$.

Result: an input-output deviation pair ϵ_S, ϵ_Y

sample $[\mathbf{E}]_{i,j} \sim \mathcal{N}(0, n_t^2/400N)$ i.i.d.

$\mathbf{E} \leftarrow (\mathbf{E} + \mathbf{E}^\top)/2$

$\mathbf{Y} \leftarrow \Phi(\mathbf{X}, \mathbf{S})$

$\hat{\mathbf{S}} \leftarrow \text{nil}$

for $t \in [T]$ **do**

$\mathbf{g} \leftarrow \nabla_{\mathbf{E}} (NF_L)^{-1} \|\Phi(\mathbf{X}, \mathbf{S} + \mathbf{E}) - \mathbf{Y}\|_F^2$

$\mathbf{g} \leftarrow (\mathbf{g} + \mathbf{g}^\top)/2$

$\mathbf{E} \leftarrow \mathbf{E} + \alpha \mathbf{g}$

$\mathbf{E} \leftarrow \min\{n_t, \|\mathbf{E}\|_2\} \mathbf{E} / \|\mathbf{E}\|_2$

$\hat{\mathbf{S}} \leftarrow \mathbf{S} + \mathbf{E}$

$\hat{\mathbf{S}} \leftarrow \min\{1, \|\hat{\mathbf{S}}\|_2\} \hat{\mathbf{S}} / \|\hat{\mathbf{S}}\|_2$

$\mathbf{E} \leftarrow \hat{\mathbf{S}} - \mathbf{S}$

end

$\epsilon_S \leftarrow \|\mathbf{E}\|_2$

$\epsilon_Y \leftarrow \|\Phi(\mathbf{X}, \hat{\mathbf{S}}) - \mathbf{Y}\|_F$

ReLU, $\sigma(\alpha x) = \alpha \sigma(x)$, $\forall x \in \mathbb{R}$, and each layer function is linear in $\mathbf{X}$. Therefore,

$$\begin{aligned} \hat{\Phi}(\mathbf{X}, \mathbf{S}) &= \hat{f}^{(L)}(\dots \sigma[\hat{f}^{(1)}(\mathbf{X}, \mathbf{S})] \dots) \\ &= \hat{f}^{(L)}(\dots \sigma[\alpha f^{(1)}(\mathbf{X}, \mathbf{S})] \dots) \\ &= \hat{f}^{(L)}(\dots \hat{f}^{(2)}(\alpha \sigma[f^{(1)}(\mathbf{X}, \mathbf{S})], \mathbf{S}) \dots) \\ &= \hat{f}^{(L)}(\dots \sigma[\alpha^2 f^{(2)}(\sigma[f^{(1)}(\mathbf{X}, \mathbf{S})], \mathbf{S})] \dots) \\ &= \dots \\ &= \alpha^L \Phi(\mathbf{X}, \mathbf{S}). \end{aligned}$$

The scaling of the signal and structure Lipschitz constants follows by the fact $\sup(\alpha S) = \alpha \sup S$ for any set S .

S7 PROOF OF PROPOSITION 4

Consider a layer function f and a version of it $\hat{f}$ with all convolutional weights Θ_k scaled by α . We have $\hat{f}(\mathbf{X}, \mathbf{S}) = \sum_{k \in \mathcal{K}} \mathbf{S}^k \mathbf{X}(\alpha \Theta_k) = \alpha f(\mathbf{X}, \mathbf{S})$. Next, observe that with